

- 1 (a) A network of resistors, each of resistance R , is shown in Fig. 7.1.

9702/22/M/J/09/Q7

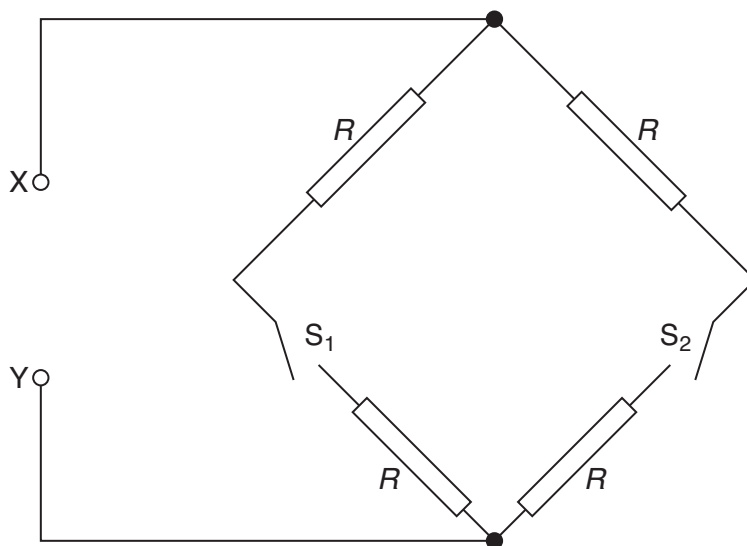


Fig. 7.1

Switches S_1 and S_2 may be 'open' or 'closed'.

Complete Fig. 7.2 by calculating the resistance, in terms of R , between points X and Y for the switches in the positions shown.

switch S_1	switch S_2	resistance between points X and Y
open	open	
open	closed	
closed	closed	

Fig. 7.2

[3]

- (b) Two cells of e.m.f. E_1 and E_2 and negligible internal resistance are connected into a network of resistors, as shown in Fig. 7.3.

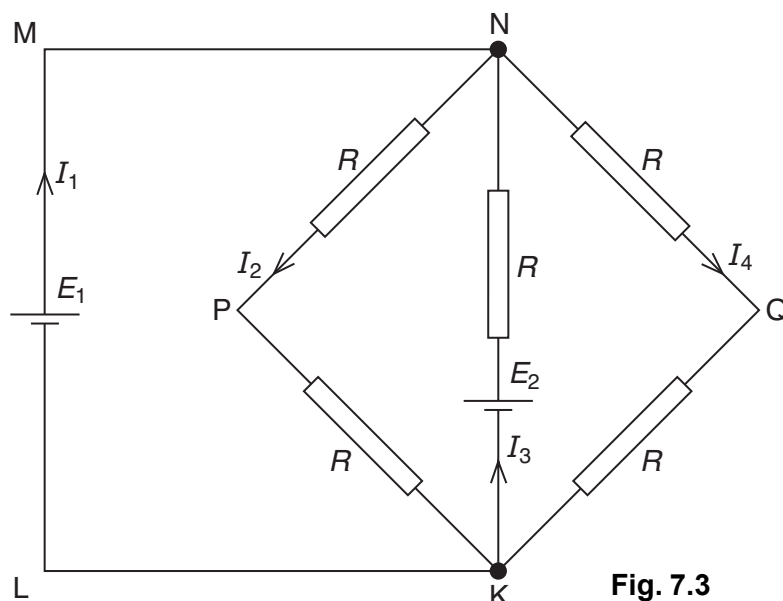


Fig. 7.3

The currents in the network are as indicated in Fig. 7.3.

Use Kirchhoff's laws to state the relation

- (i) between currents I_1 , I_2 , I_3 and I_4 ,

.....[1]

- (ii) between E_1 , E_2 , R , and I_3 in loop NKLMN,

.....[1]

- (iii) between E_2 , R , I_3 and I_4 in loop NKQN.

.....[1]

9702/21/O/N/09/Q6

- 2 A cell has electromotive force (e.m.f.) E and internal resistance r . It is connected in series with a variable resistor R , as shown in Fig. 6.1.

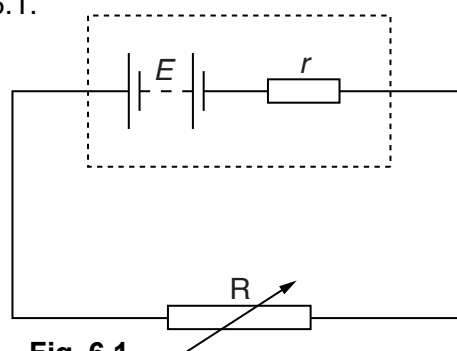


Fig. 6.1

- (a) Define electromotive force (e.m.f.).

.....

 [2]

(b) The variable resistor R has resistance X. Show that

$$\frac{\text{power dissipated in resistor R}}{\text{power produced in cell}} = \frac{X}{X + r}.$$

[3]

(c) The variation with resistance X of the power P_R dissipated in R is shown in Fig. 6.2.

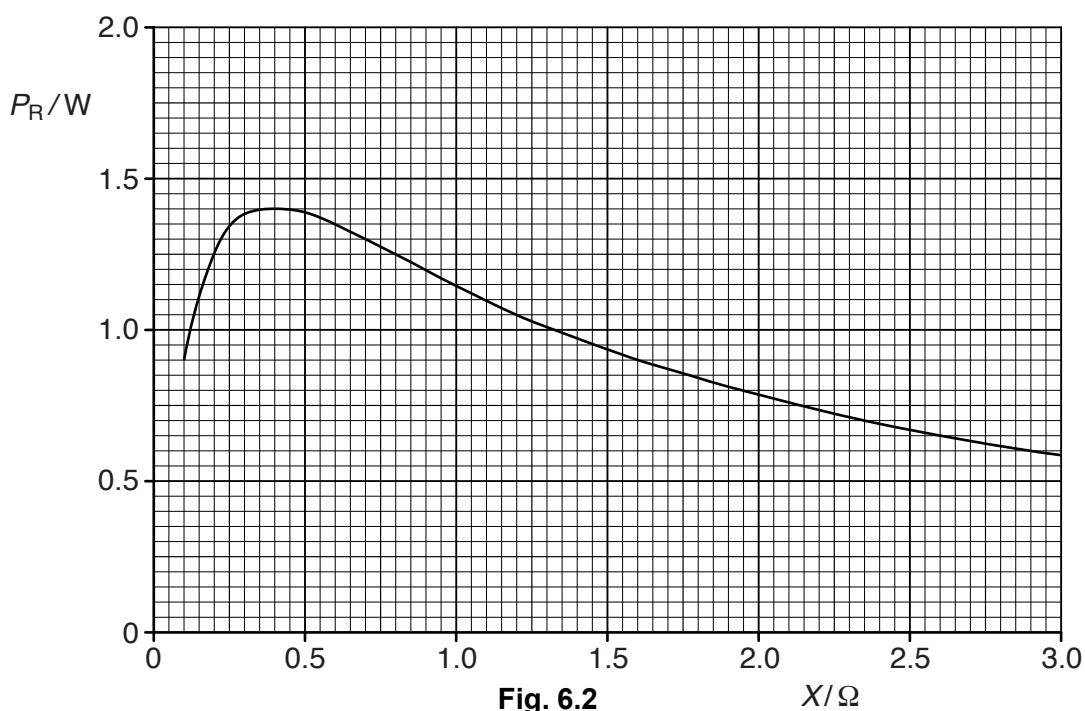


Fig. 6.2

X/Ω

(i) Use Fig. 6.2 to state, for maximum power dissipation in resistor R, the magnitude of this power and the resistance of R.

maximum power = W

resistance = Ω

[2]

(ii) The cell has e.m.f. 1.5V.

Use your answers in (i) to calculate the internal resistance of the cell.

internal resistance = Ω [3]

- (d) In Fig. 6.2, it can be seen that, for larger values of X , the power dissipation decreases. Use the relationship in (b) to suggest one advantage, despite the lower power output, of using the cell in a circuit where the resistance X is larger than the internal resistance of the cell.

.....
 [1]

9702/22/O/N/09/Q6

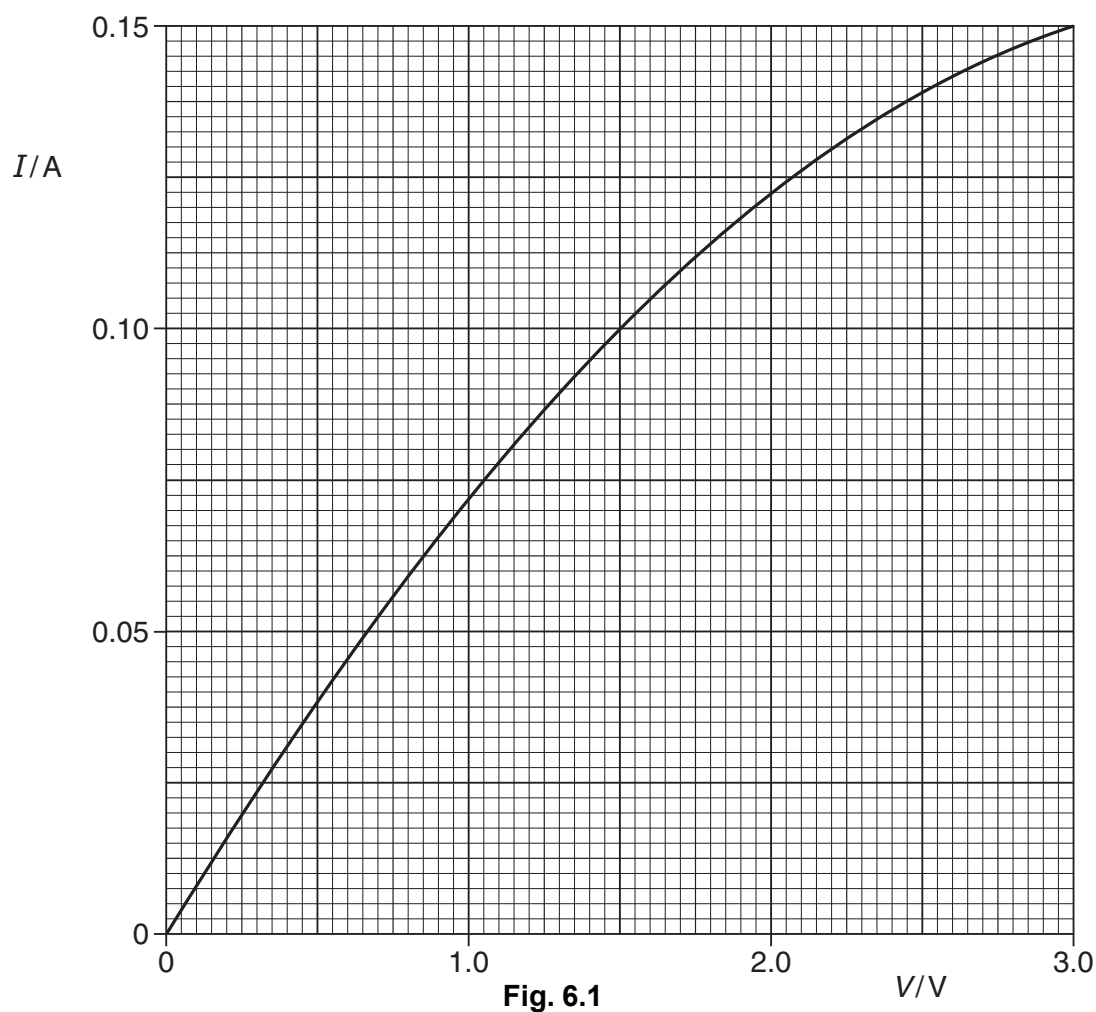
- 3 (a) Two resistors, each of resistance R , are connected first in series and then in parallel. Show that the ratio

$$\frac{\text{combined resistance of resistors connected in series}}{\text{combined resistance of resistors connected in parallel}}$$

is equal to 4.

[1]

- (b) The variation with potential difference V of the current I in a lamp is shown in Fig. 6.1.



Calculate the resistance of the lamp for a potential difference across the lamp of 1.5V.

resistance = Ω [2]

- (c) Two lamps, each having the I - V characteristic shown in Fig. 6.1, are connected first in series and then in parallel with a battery of e.m.f. 3.0V and negligible internal resistance.

Complete the table of Fig. 6.2 for the lamps connected to the battery.

	p.d. across each lamp/V	resistance of each lamp/ Ω	combined resistance of lamps/ Ω
lamps connected in series			
lamps connected in parallel			

Fig. 6.2

[4]

- (d) (i) Use data from the completed Fig. 6.2 to calculate the ratio

$$\frac{\text{combined resistance of lamps connected in series}}{\text{combined resistance of lamps connected in parallel}}$$

ratio = [1]

- (ii) The ratios in (a) and (d)(i) are not equal.

By reference to Fig. 6.1, state and explain qualitatively the change in the resistance of a lamp as the potential difference is changed.

.....

 [3]

- 4 (a) A metal wire of constant resistance is used in an electric heater.
In order not to overload the circuit for the heater, the supply voltage to the heater is reduced from 230 V to 220 V.

Determine the percentage reduction in the power output of the heater.

reduction = % [2]

- (b) A uniform wire AB of length 100 cm is connected between the terminals of a cell of e.m.f. 1.5 V and negligible internal resistance, as shown in Fig. 6.1.

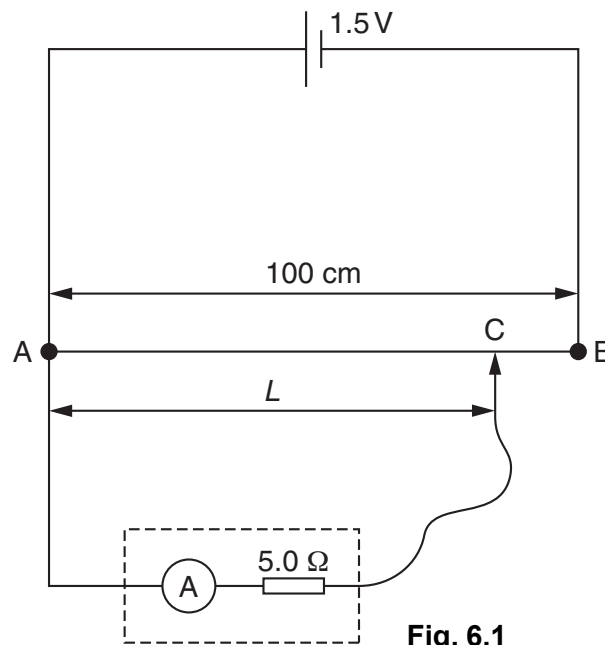


Fig. 6.1

An ammeter of internal resistance $5.0\ \Omega$ is connected to end A of the wire and to a contact C that can be moved along the wire.

Determine the reading on the ammeter for the contact C placed

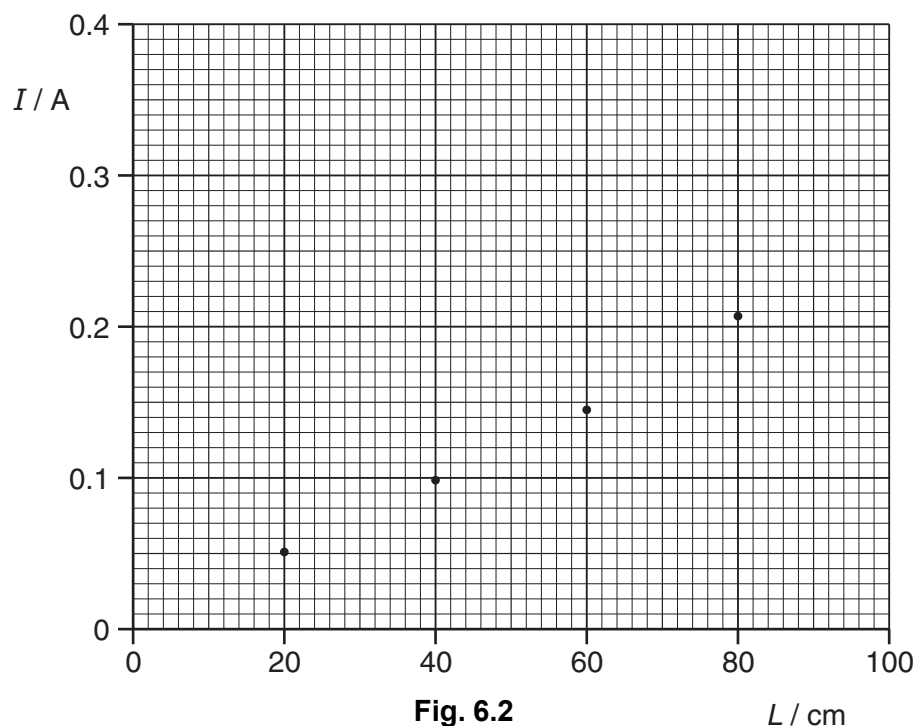
- (i) at A,

reading = A [1]

- (ii) at B.

reading = A [1]

- (c) Using the circuit in (b), the ammeter reading I is recorded for different distances L of the contact C from end A of the wire. Some data points are shown on Fig. 6.2.



- (i) Use your answers in (b) to plot data points on Fig. 6.2 corresponding to the contact C placed at end A and at end B of the wire.

[1]

- (ii) Draw a line of best fit for all of the data points and hence determine the ammeter reading for contact C placed at the midpoint of the wire.

reading = A [1]

- (iii) Use your answer in (ii) to calculate the potential difference between A and the contact C for the contact placed at the midpoint of AB.

potential difference = V [2]

- (d) Explain why, although the contact C is at the midpoint of wire AB, the answer in (c)(iii) is **not** numerically equal to one half of the e.m.f. of the cell.

.....
 [2]

5 (a) A lamp is rated as 12V, 36W.

(i) Calculate the resistance of the lamp at its working temperature.

resistance = Ω [2]

(ii) On the axes of Fig. 6.1, sketch a graph to show the current-voltage (I – V) characteristic of the lamp. Mark an appropriate scale for current on the y -axis.

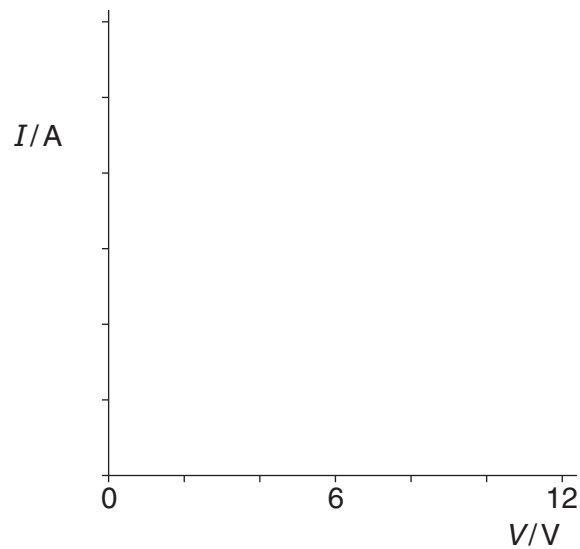


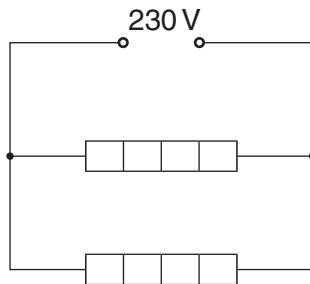
Fig. 6.1

[3]

(b) Some heaters are each labelled 230V, 1.0kW. The heaters have constant resistance.

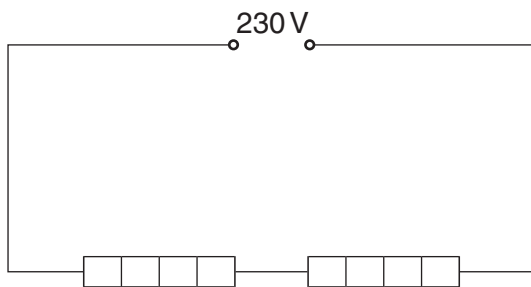
Determine the total power dissipation for the heaters connected as shown in each of the diagrams shown below.

(i)



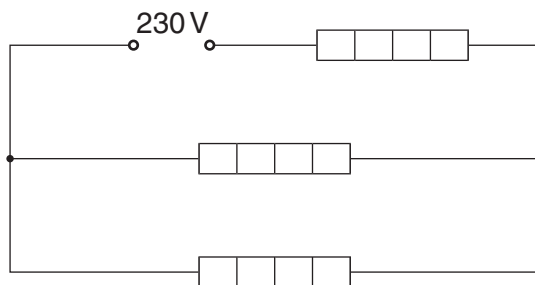
power = kW [1]

(ii)



power = kW [1]

(iii)



power = kW [2]

- 6 (a) A variable resistor is used to control the current in a circuit, as shown in Fig. 5.1.

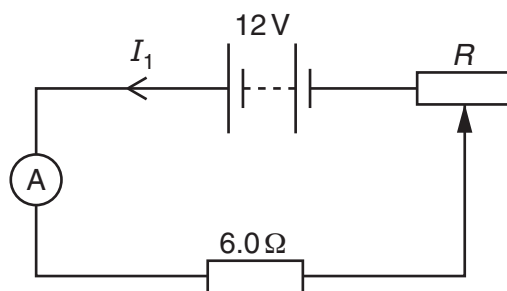


Fig. 5.1

The variable resistor is connected in series with a 12V power supply of negligible internal resistance, an ammeter and a 6.0Ω resistor. The resistance R of the variable resistor can be varied between 0 and 12Ω .

- (i) The maximum possible current in the circuit is 2.0A. Calculate the minimum possible current.

minimum current = A [2]

- (ii) On Fig. 5.2, sketch the variation with R of current I_1 in the circuit.

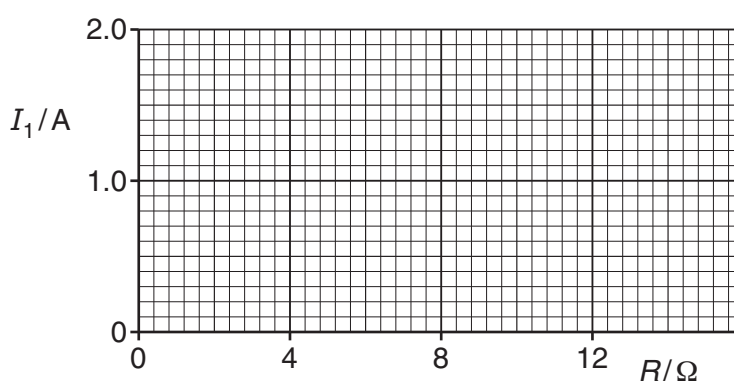


Fig. 5.2

[2]

(b) The variable resistor in (a) is now connected as a potential divider, as shown in Fig. 5.3.

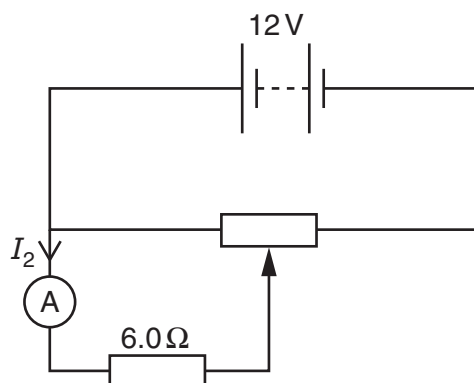


Fig. 5.3

Calculate the maximum possible and minimum possible current I_2 in the ammeter.

maximum $I_2 = \dots\dots\dots$ A

minimum $I_2 = \dots\dots\dots$ A
[2]

(c) (i) Sketch on Fig. 5.4 the $I - V$ characteristic of a filament lamp.

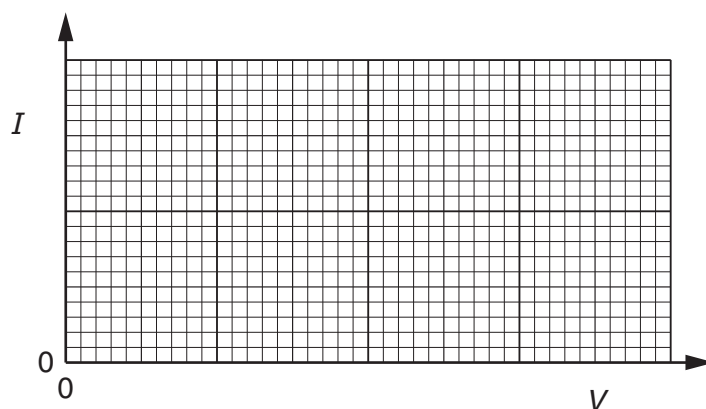


Fig. 5.4

[2]

(ii) The resistor of resistance 6.0Ω is replaced with a filament lamp in the circuits of Fig. 5.1 and Fig. 5.3. State an advantage of using the circuit of Fig. 5.3, compared to the circuit of Fig 5.1, when using the circuits to vary the brightness of the filament lamp.

.....
..... [1]

7 (a) For a cell, explain the terms

9702/22/M/J/11/Q5

(i) *electromotive force (e.m.f.),*

.....
..... [1]

(ii) *internal resistance.*

.....
..... [1]

(b) The circuit of Fig. 5.1 shows two batteries A and B and a resistor R connected in series.

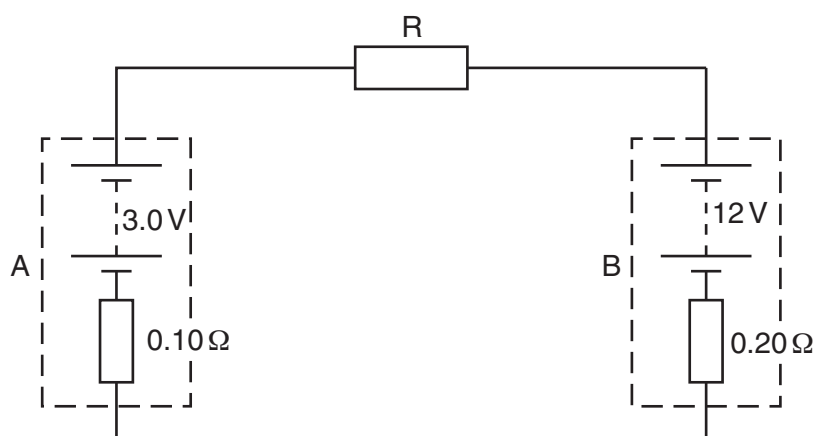


Fig. 5.1

Battery A has an e.m.f. of 3.0V and an internal resistance of 0.10Ω . Battery B has an e.m.f. of 12V and an internal resistance of 0.20Ω . Resistor R has a resistance of 3.3Ω .

- (i) Apply Kirchhoff's second law to calculate the current in the circuit.

current = A [2]

- (ii) Calculate the power transformed by battery B.

power = W [2]

- (iii) Calculate the total energy lost per second in resistor R and the internal resistances.

energy lost per second = Js^{-1} [2]

- (c) The circuit of Fig. 5.1 may be used to store energy in battery A. Suggest how your answers in (b) support this statement.

.....
..... [1]

- 8 A potentiometer circuit that is used as a means of comparing potential differences is shown in Fig. 5.1.

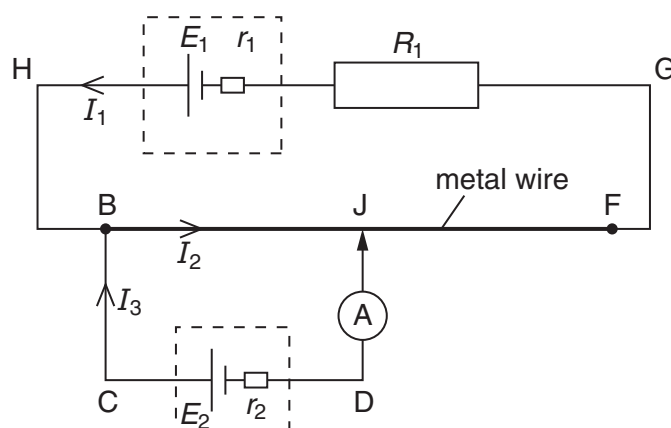


Fig. 5.1

A cell of e.m.f. E_1 and internal resistance r_1 is connected in series with a resistor of resistance R_1 and a uniform metal wire of total resistance R_2 .

A second cell of e.m.f. E_2 and internal resistance r_2 is connected in series with a sensitive ammeter and is then connected across the wire at BJ. The connection at J is halfway along the wire. The current directions are shown on Fig. 5.1.

- (a) Use Kirchhoff's laws to obtain the relation

- (i) between the currents I_1 , I_2 and I_3 ,

.....[1]

- (ii) between E_1 , R_1 , R_2 , r_1 , I_1 and I_2 in loop HBJFGH,

.....[1]

- (iii) between E_1 , E_2 , r_1 , r_2 , R_1 , R_2 , I_1 and I_3 in the loop HBCDJFGH.

.....[2]

- (b) The connection at J is moved along the wire. Explain why the reading on the ammeter changes.

.....

.....

.....

.....[2]

9 (a) (i) State Kirchhoff's second law.

9702/21/M/J/12/Q5

.....
[1]

(ii) Kirchhoff's second law is linked to the conservation of a certain quantity. State this quantity.

.....[1]

(b) The circuit shown in Fig. 5.1 is used to compare potential differences.

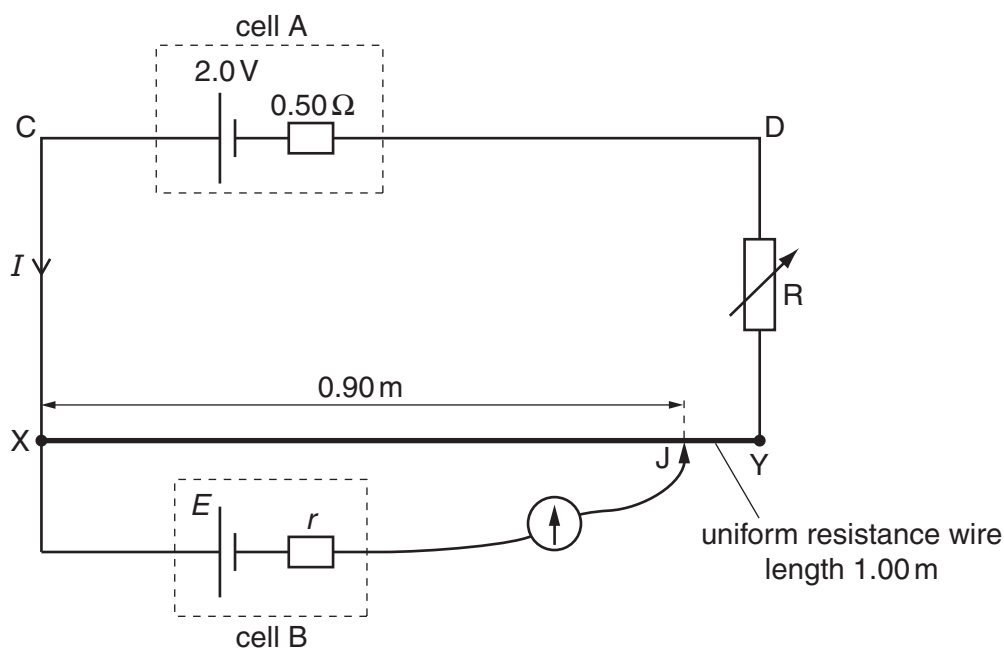


Fig. 5.1

The uniform resistance wire XY has length 1.00m and resistance 4.0Ω . Cell A has e.m.f. 2.0V and internal resistance 0.50Ω . The current through cell A is I . Cell B has e.m.f. E and internal resistance r .

The current through cell B is made zero when the movable connection J is adjusted so that the length of XJ is 0.90m. The variable resistor R has resistance 2.5Ω .

(i) Apply Kirchhoff's second law to the circuit CXYDC to determine the current I .

$I =$ A [2]

- (ii) Calculate the potential difference across the length of wire XJ.

potential difference = V [2]

- (iii) Use your answer in (ii) to state the value of E .

$E =$ V [1]

- (iv) State why the value of the internal resistance of cell B is not required for the determination of E .

.....

.....[1]

- 10 A battery of electromotive force 12V and negligible internal resistance is connected to two resistors and a light-dependent resistor (LDR), as shown in Fig. 4.1.

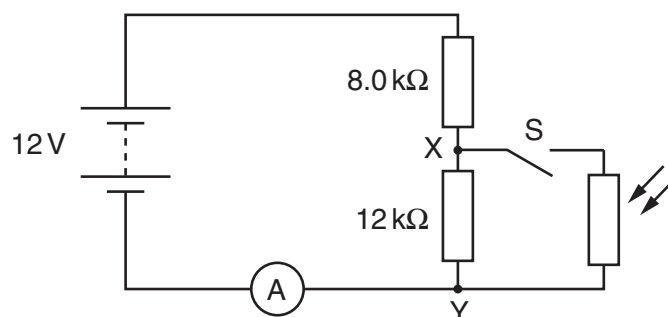


Fig. 4.1

An ammeter is connected in series with the battery. The LDR and switch S are connected across the points XY.

- (a) The switch S is open. Calculate the potential difference (p.d.) across XY.

p. d. = V [3]

- (b) The switch S is closed. The resistance of the LDR is $4.0\text{ k}\Omega$. Calculate the current in the ammeter.

current = A [3]

- (c) The switch S remains closed. The intensity of the light on the LDR is increased. State and explain the change to

- (i) the ammeter reading,

.....
 [2]

- (ii) the p.d. across XY.

.....
 [2]

11 (a) (i) State Kirchhoff's first law.

9702/23/M/J/12/Q5

.....[1]

(ii) Kirchhoff's first law is linked to the conservation of a certain quantity. State this quantity.

.....[1]

(b) A variable resistor of resistance R is used to control the current in a circuit, as shown in Fig. 5.1.

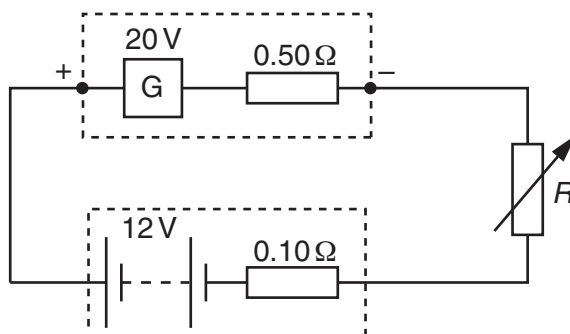


Fig. 5.1

The generator G has e.m.f. 20 V and internal resistance 0.50Ω . The battery has e.m.f. 12 V and internal resistance 0.10Ω . The current in the circuit is 2.0 A .

(i) Apply Kirchhoff's second law to the circuit to determine the resistance R .

$R = \dots\dots\dots \Omega$ [2]

(ii) Calculate the total power generated by G .

power = $\dots\dots\dots \text{ W}$ [2]

(iii) Calculate the power loss in the total resistance of the circuit.

power = $\dots\dots\dots \text{ W}$ [2]

(iv) The circuit is used to supply energy to the battery from the generator. Determine the efficiency of the circuit.

efficiency = $\dots\dots\dots$ [2]

- 12 Fig. 5.1 shows a 12V power supply with negligible internal resistance connected to a uniform metal wire AB. The wire has length 1.00 m and resistance $10\ \Omega$. Two resistors of resistance $4.0\ \Omega$ and $2.0\ \Omega$ are connected in series across the wire.

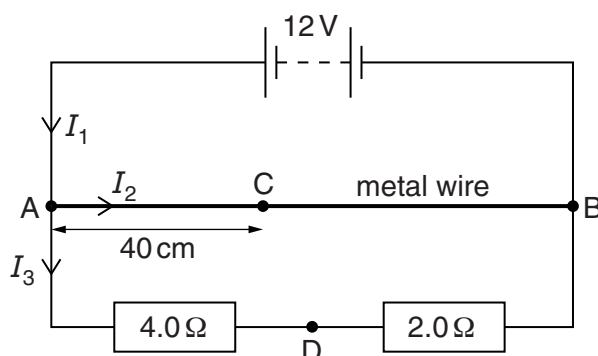


Fig. 5.1

Currents I_1 , I_2 and I_3 in the circuit are as shown in Fig. 5.1.

- (a) (i) Use Kirchhoff's first law to state a relationship between I_1 , I_2 and I_3 .

..... [1]

- (ii) Calculate I_1 .

$I_1 = \dots\dots\dots$ A [3]

- (iii) Calculate the ratio x , where

$$x = \frac{\text{power in metal wire}}{\text{power in series resistors}}.$$

$x = \dots\dots\dots$ [3]

- (b) Calculate the potential difference (p.d.) between the points C and D, as shown in Fig. 5.1. The distance AC is 40 cm and D is the point between the two series resistors.

p.d. = V [3]

- 13 A circuit used to measure the power transfer from a battery is shown in Fig. 4.1. The power is transferred to a variable resistor of resistance R .

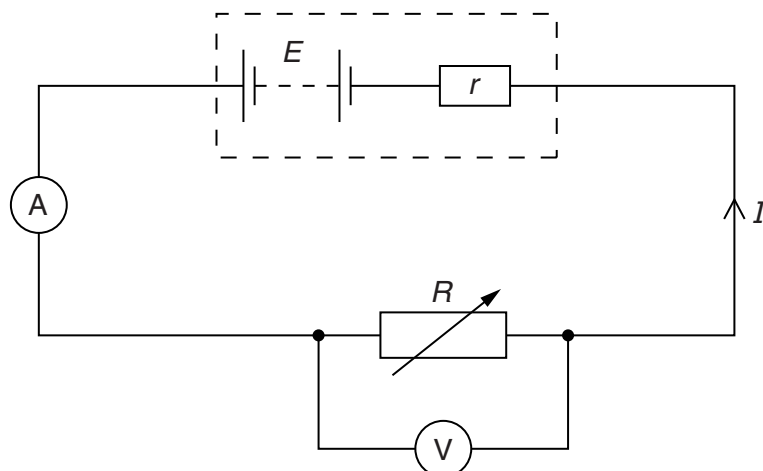


Fig. 4.1

The battery has an electromotive force (e.m.f.) E and an internal resistance r . There is a potential difference (p.d.) V across R . The current in the circuit is I .

- (a) By reference to the circuit shown in Fig. 4.1, distinguish between the definitions of e.m.f. and p.d.

.....

.....

.....

..... [3]

- (b) Using Kirchhoff's second law, determine an expression for the current I in the circuit.

(c) The variation with current I of the p.d. V across R is shown in Fig. 4.2.

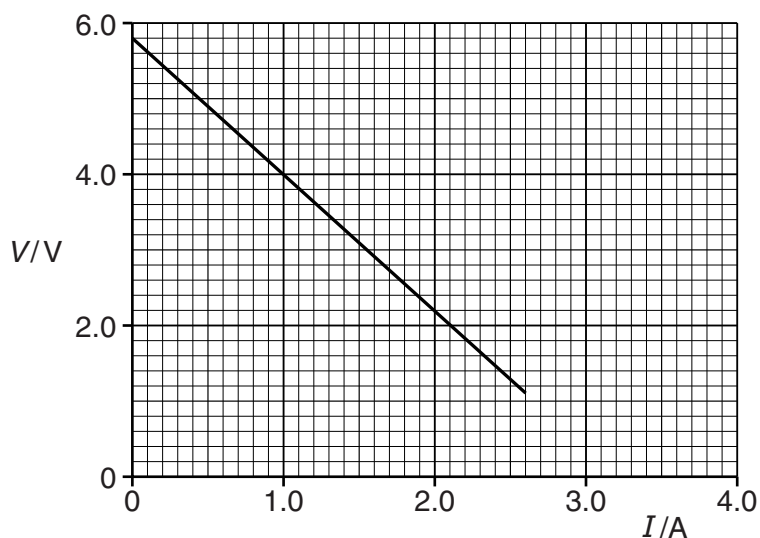


Fig. 4.2

Use Fig. 4.2 to determine

(i) the e.m.f. E ,

$E = \dots\dots\dots$ V [1]

(ii) the internal resistance r .

$r = \dots\dots\dots$ Ω [2]

(d) (i) Using data from Fig. 4.2, calculate the power transferred to R for a current of 1.6 A.

power = $\dots\dots\dots$ W [2]

(ii) Use your answers from (c)(i) and (d)(i) to calculate the efficiency of the battery for a current of 1.6 A.

efficiency = $\dots\dots\dots$ % [2]

14 (a) Define *potential difference* (p.d.).

9702/23/M/J/13/Q6

..... [1]

- (b) A battery of electromotive force 20V and zero internal resistance is connected in series with two resistors R_1 and R_2 , as shown in Fig. 6.1.

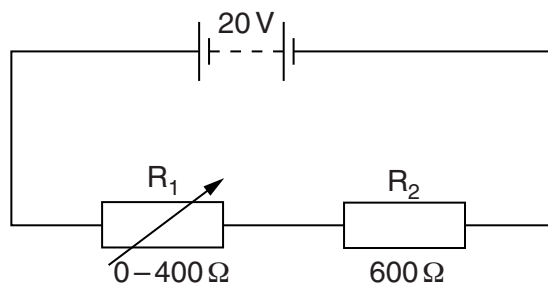


Fig. 6.1

The resistance of R_2 is 600 Ω . The resistance of R_1 is varied from 0 to 400 Ω .

Calculate

- (i) the maximum p.d. across R_2 ,

maximum p.d. = V [1]

- (ii) the minimum p.d. across R_2 .

minimum p.d. = V [2]

- (c) A light-dependent resistor (LDR) is connected in parallel with R_2 , as shown in Fig. 6.2.

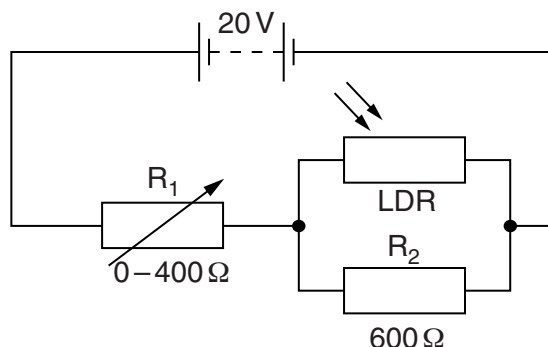


Fig. 6.2

When the light intensity is varied, the resistance of the LDR changes from $5.0\text{k}\Omega$ to $1.2\text{k}\Omega$.

- (i) For the **maximum** light intensity, calculate the total resistance of R_2 and the LDR.

total resistance = Ω [2]

- (ii) The resistance of R_1 is varied from 0 to 400Ω in the circuits of Fig. 6.1 and Fig. 6.2. State and explain the difference, if any, between the minimum p.d. across R_2 in each circuit. Numerical values are not required.

.....

.....

.....[2]

15 A battery is connected in series with resistors X and Y, as shown in Fig. 6.1.

9702/22/M/J/14/Q6

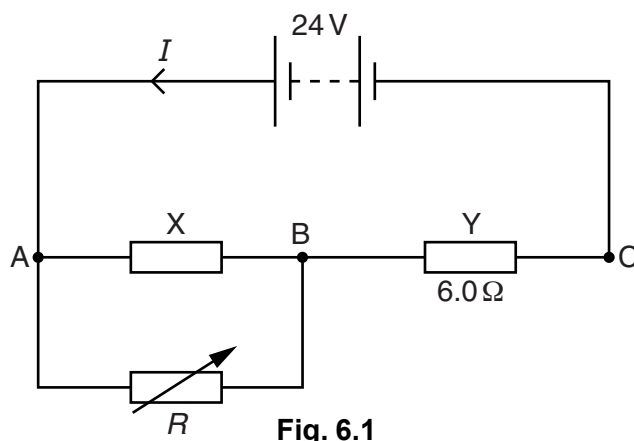


Fig. 6.1

The resistance of X is constant. The resistance of Y is $6.0\ \Omega$. The battery has electromotive force (e.m.f.) 24V and zero internal resistance. A variable resistor of resistance R is connected in parallel with X.

The current I from the battery is changed by varying R from $5.0\ \Omega$ to $20\ \Omega$. The variation with R of I is shown in Fig. 6.2.

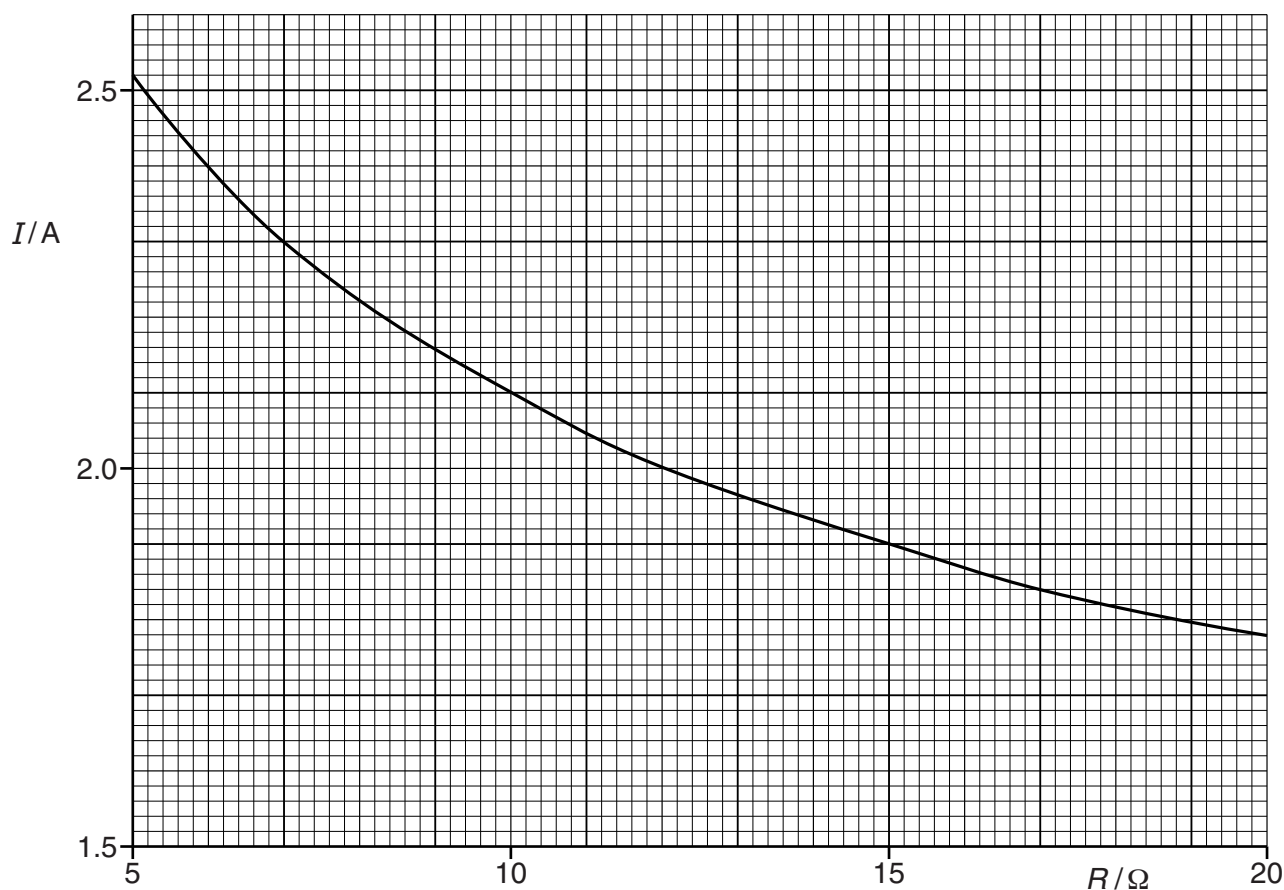


Fig. 6.2

(a) Explain why the potential difference (p.d.) between points A and C is 24V for all values of R .

.....
 [1]

- (b) Use Fig. 6.2 to state and explain the variation of the p.d. across resistor Y as R is increased. Numerical values are not required.

.....
.....
..... [2]

- (c) For $R = 6.0\Omega$,

- (i) show that the p.d. between points A and B is 9.6V,

[2]

- (ii) calculate the resistance of X,

resistance = Ω [3]

- (iii) calculate the power provided by the battery.

power = W [2]

- (d) State and explain qualitatively how the power provided by the battery changes as the resistance R is increased.

.....
..... [1]

- 16** A battery of electromotive force (e.m.f.) 12V and internal resistance r is connected in series to two resistors, each of constant resistance X , as shown in Fig. 5.1.

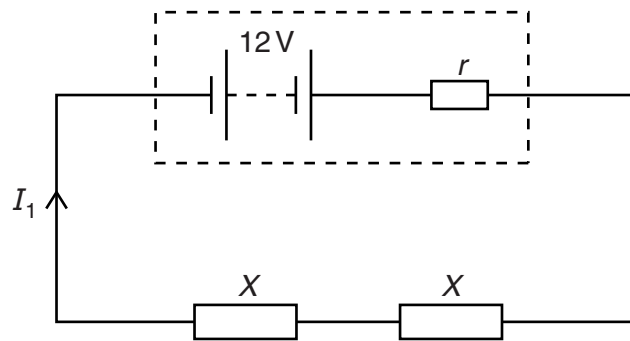


Fig. 5.1

The current I_1 supplied by the battery is 1.2 A.

The same battery is now connected to the same two resistors in parallel, as shown in Fig. 5.2.

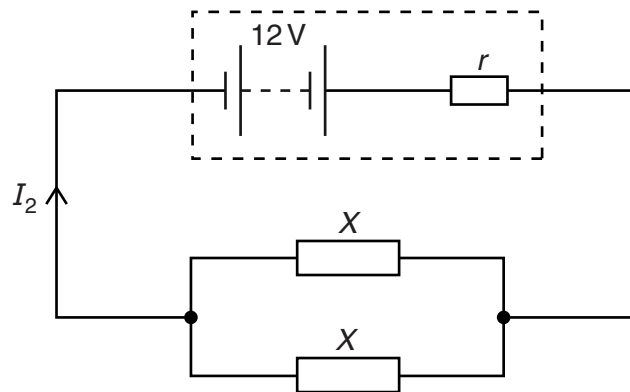


Fig. 5.2

The current I_2 supplied by the battery is 3.0 A.

- (a) (i)** Show that the combined resistance of the two resistors, each of resistance X , is four times greater in Fig. 5.1 than in Fig. 5.2.

[2]

- (ii)** Explain why I_2 is not four times greater than I_1 .

.....

.....

.....[2]

(iii) Using Kirchhoff's second law, state equations, in terms of e.m.f., current, X and r , for

1. the circuit of Fig. 5.1,

.....

2. the circuit of Fig. 5.2.

.....

[2]

(iv) Use the equations in (iii) to calculate the resistance X .

$X = \dots\dots\dots \Omega$ [1]

(b) Calculate the ratio

$$\frac{\text{power transformed in one resistor of resistance } X \text{ in Fig. 5.1}}{\text{power transformed in one resistor of resistance } X \text{ in Fig. 5.2}}$$

ratio =[2]

(c) The resistors in Fig. 5.1 and Fig. 5.2 are replaced by identical 12V filament lamps.

Explain why the resistance of each lamp, when connected in series, is not the same as the resistance of each lamp when connected in parallel.

.....

.....

.....

.....[2]

- 17 (a)** A wire has length 100cm and diameter 0.38mm. The metal of the wire has resistivity $4.5 \times 10^{-7} \Omega \text{m}$.

Show that the resistance of the wire is 4.0Ω .

[3]

- (b)** The ends B and D of the wire in **(a)** are connected to a cell X, as shown in Fig. 6.1.

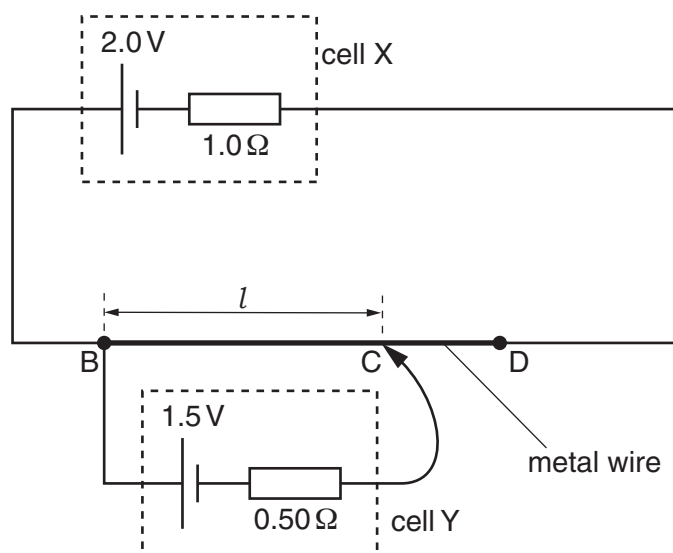


Fig. 6.1

The cell X has electromotive force (e.m.f.) 2.0V and internal resistance 1.0Ω .

A cell Y of e.m.f. 1.5V and internal resistance 0.50Ω is connected to the wire at points B and C, as shown in Fig. 6.1.

The point C is distance l from point B. The current in cell Y is zero.

Calculate

- (i)** the current in cell X,

current = A [2]

(ii) the potential difference (p.d.) across the wire BD,

p.d. = V [1]

(iii) the distance l .

l = cm [2]

(c) The connection at C is moved so that l is increased. Explain why the e.m.f. of cell Y is less than its terminal p.d.

.....
.....
.....[2]

9702/22/M/J/15/Q5

18 (a) On Fig. 5.1, sketch the temperature characteristic of a thermistor.

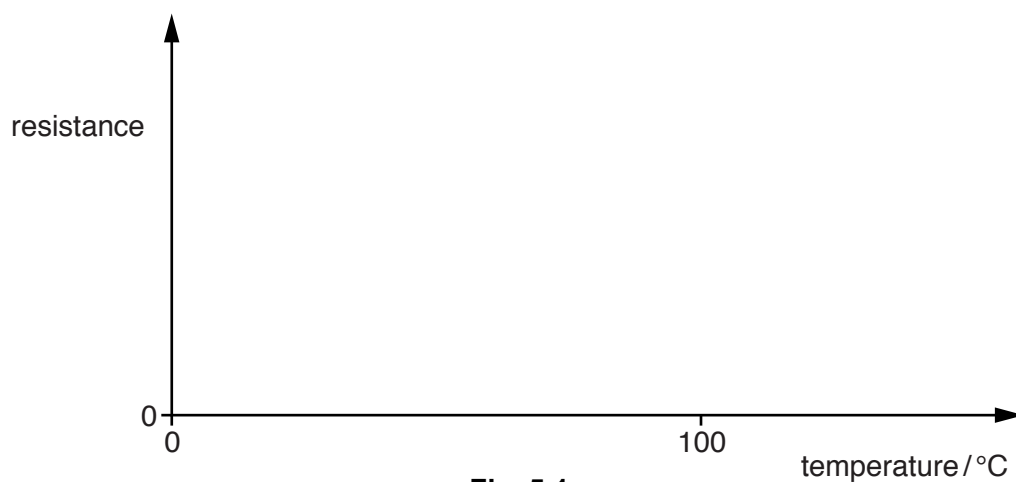
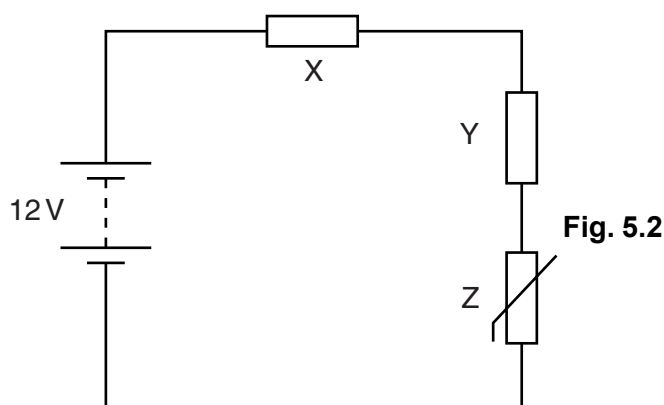


Fig. 5.1

[2]

(b) A potential divider circuit is shown in Fig. 5.2.



The battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected in series with resistors X and Y and thermistor Z. The resistance of Y is $15\text{ k}\Omega$ and the resistance of Z at a particular temperature is $3.0\text{ k}\Omega$. The potential difference (p.d.) across Y is 8.0 V.

- (i) Explain why the power transformed in the battery equals the total power transformed in X, Y and Z.

..... [1]

- (ii) Calculate the current in the circuit.

current = A [2]

- (iii) Calculate the resistance of X.

resistance = Ω [3]

- (iv) The temperature of Z is increased.

State and explain the effect on the potential difference across Z.

.....

 [2]

- 19** A uniform resistance wire AB has length 50cm and diameter 0.36mm. The resistivity of the metal of the wire is $5.1 \times 10^{-7} \Omega \text{ m}$.

(a) Show that the resistance of the wire AB is 2.5Ω .

[2]

- (b) The wire AB is connected in series with a power supply E and a resistor R as shown in Fig. 5.1.

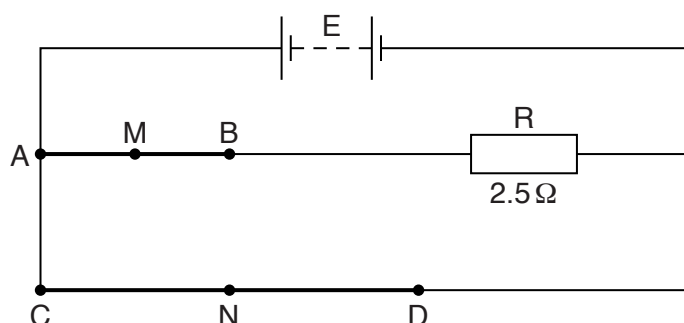


Fig. 5.1

The electromotive force (e.m.f.) of E is 6.0V and its internal resistance is negligible. The resistance of R is 2.5Ω . A second uniform wire CD is connected across the terminals of E. The wire CD has length 100cm, diameter 0.18mm and is made of the same metal as wire AB.

Calculate

- (i) the current supplied by E,

current = A [4]

(ii) the power transformed in wire AB,

power = W [2]

(iii) the potential difference (p.d.) between the midpoint M of wire AB and the midpoint N of wire CD.

p.d. = V [2]

20 (a) Define electric *potential difference* (p.d.).

.....
[1]

(b) A battery of electromotive force (e.m.f.) 14V and negligible internal resistance is connected to a resistor network, as shown in Fig. 6.1.

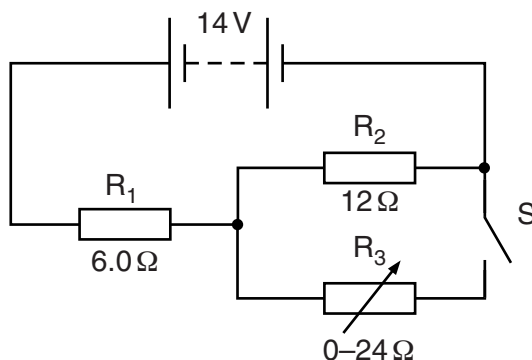


Fig. 6.1

R_1 and R_2 are fixed resistors of resistances 6.0Ω and 12Ω respectively. R_3 is a variable resistor.

Switch S is **closed**.

(i) Calculate the current in the battery when the resistance of R_3 is set

1. at zero,

current = A [2]

2. at 24Ω .

current = A [2]

- (ii) Use your answers in (b)(i) to calculate the change in the total power produced by the battery when the resistance of R_3 is changed from zero to $24\ \Omega$.

change in power = W [2]

- (c) Switch S in Fig. 6.1 is now **opened**.

Resistors R_1 and R_2 are made from metal wires. Some data for these resistors are shown in Fig. 6.2.

	R_1	R_2
cross-sectional area of wire	A	$1.8A$
number of free electrons per unit volume in metal	n	$0.50n$

Fig. 6.2

Determine the ratio

$$\frac{\text{average drift speed of free electrons in } R_1}{\text{average drift speed of free electrons in } R_2}.$$

ratio = [2]

21 (a) State Kirchhoff's second law.

.....

.....

.....

.....[2]

(b) A battery is connected in parallel with two lamps A and B, as shown in Fig. 5.1.

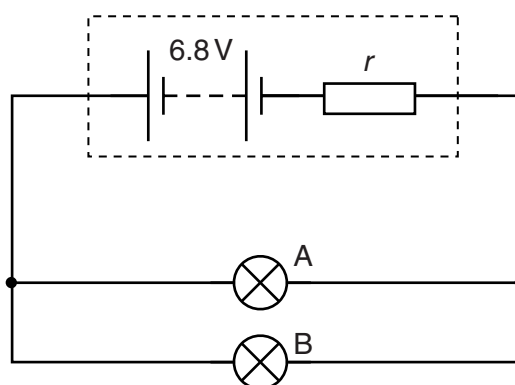


Fig. 5.1

The battery has electromotive force (e.m.f.) 6.8V and internal resistance r .

The I - V characteristics of lamps A and B are shown in Fig. 5.2.

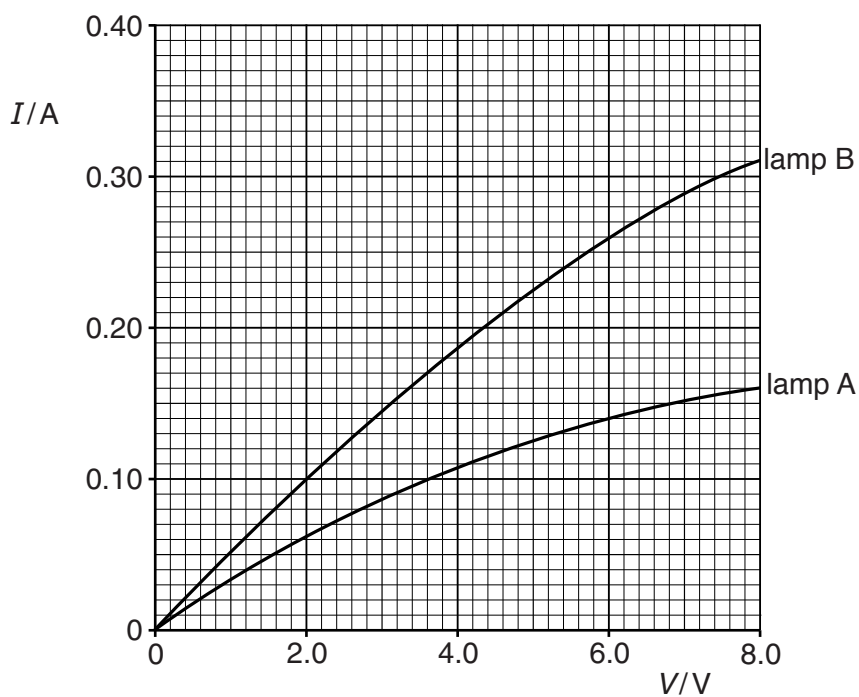


Fig. 5.2

The potential difference across the battery terminals is 6.0V.

- (i) Use Fig. 5.2 to show that the current in the battery is 0.40 A.

[2]

- (ii) Calculate the internal resistance r of the battery.

$r = \dots\dots\dots \Omega$ [2]

- (iii) Determine the ratio

$$\frac{\text{resistance of lamp A}}{\text{resistance of lamp B}}.$$

ratio = $\dots\dots\dots$ [2]

(iv) Determine

1. the total power produced by the battery,

power = W [2]

2. the efficiency of the battery in the circuit.

efficiency = [2]

22 (a) Define the *ohm*.

9702/21/M/J/17/Q6

.....
[1]

- (b) A cell X of electromotive force (e.m.f.) 1.5 V and negligible internal resistance is connected in series to three resistors A, B and C, as shown in Fig. 6.1.

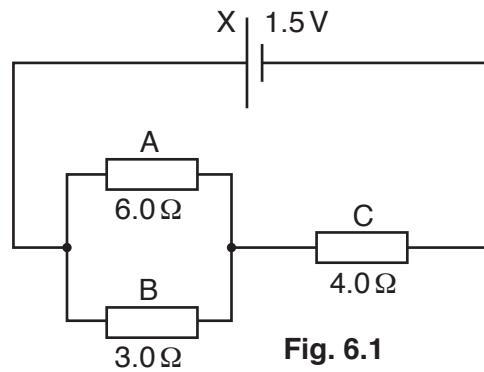


Fig. 6.1

Resistors A and B have resistances 6.0Ω and 3.0Ω respectively and are connected in parallel. Resistor C has resistance 4.0Ω and is connected in series with the parallel combination.

Calculate

- (i) the current in the circuit,

current =A [3]

- (ii) the current in resistor B,

current =A [1]

- (iii) the ratio

$$\frac{\text{power dissipated in resistor B}}{\text{power dissipated in resistor C}}$$

ratio =[2]

(c) The resistors A, B and C in (b) are wires of the same material and have the same length.

(i) Explain how the resistors may be made with different resistance values.

.....[1]

(ii) Calculate the ratio

$$\frac{\text{average drift speed of the charge carriers in resistor B}}{\text{average drift speed of the charge carriers in resistor C}}.$$

ratio =[2]

(d) A cell of e.m.f. 1.5 V and negligible internal resistance is connected in parallel with cell X in Fig. 6.1 with their positive terminals together.

State the change, if any, to the current in

(i) cell X,

.....[1]

(ii) resistor C.

.....[1]

23 (a) Define *electromotive force* (e.m.f.) of a cell.

9702/22/M/J/17/Q7

.....
.....[1]

(b) A cell C of e.m.f. 1.50 V and internal resistance $0.200\ \Omega$ is connected in series with resistors X and Y, as shown in Fig. 7.1.

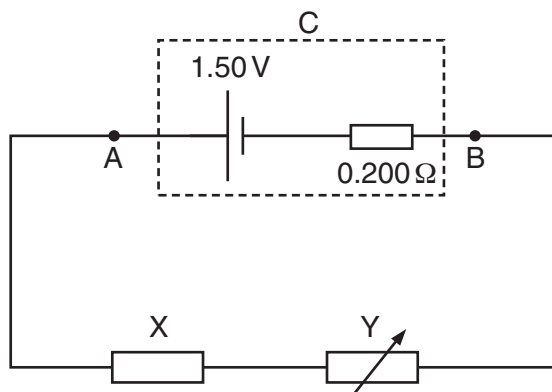


Fig. 7.1

The resistance of X is constant and the resistance of Y can be varied.

(i) The resistance of Y is varied from 0 to $8.00\ \Omega$.

State and explain the variation in the potential difference (p.d.) between points A and B (terminal p.d. across C). Numerical values are not required.

.....
.....
.....
.....[3]

(ii) The resistance of Y is set at $6.00\ \Omega$. The current in the circuit is 0.180 A .

Calculate

1. the resistance of X,

resistance = Ω [2]

2. the p.d. between points A and B,

p.d. = V [2]

3. the efficiency of the cell.

efficiency = [2]

- 24 Three cells of electromotive forces (e.m.f.) E_1 , E_2 and E_3 are connected into a circuit, as shown in Fig. 5.1.

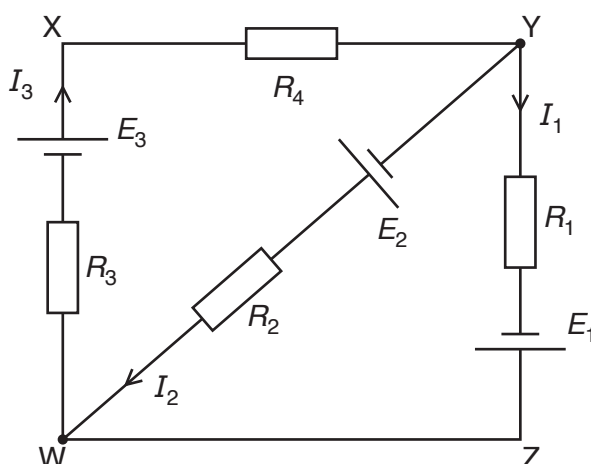


Fig. 5.1

The circuit contains resistors of resistances R_1 , R_2 , R_3 and R_4 . The currents in the different parts of the circuit are I_1 , I_2 and I_3 . The cells have negligible internal resistance.

Use Kirchhoff's laws to state an equation relating

- (a) I_1 , I_2 and I_3 ,

.....[1]

- (b) E_1 , E_3 , R_1 , R_3 , R_4 , I_1 and I_3 in loop WXYZW,

.....
[1]

- (c) E_1 , E_2 , R_1 , R_2 , I_1 and I_2 in loop YZWY.

.....
[1]

25 (a) (i) State Kirchhoff's first law.

.....
.....[1]

(ii) Kirchhoff's first law is linked to the conservation of a certain quantity. State this quantity.

.....[1]

(b) A battery of electromotive force (e.m.f.) 8.0 V and internal resistance $2.0\ \Omega$ is connected to a resistor X and a wire Y, as shown in Fig. 6.1.

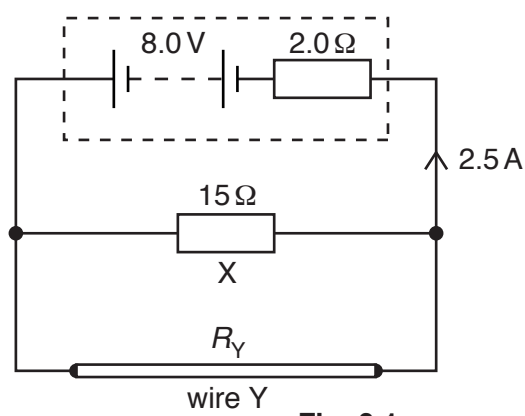


Fig. 6.1

The resistance of X is $15\ \Omega$. The resistance of Y is R_Y . The current in the battery is 2.5 A .

(i) Calculate

1. the thermal energy dissipated in the battery in a time of 5.0 minutes,

energy = J [2]

2. the terminal potential difference of the battery.

terminal potential difference = V [1]

(ii) Determine the resistance R_Y .

$R_Y = \dots\dots\dots\ \Omega$ [3]

(iii) A new wire Z has the same length but less resistance than wire Y.

1. State two possible differences between wire Z and wire Y that would separately cause wire Z to have less resistance than wire Y.

first difference:

.....

second difference:

.....

[2]

2. Wire Y is replaced in the circuit by wire Z. By considering the current in the battery, state and explain the effect of changing the wires on the total power produced by the battery.

.....

.....[2]

26. 9702/21/O/N/18/Q6

(a) State Kirchhoff's second law.

.....

 [2]

(b) An electric heater containing two heating wires X and Y is connected to a power supply of electromotive force (e.m.f.) 9.0 V and negligible internal resistance, as shown in Fig. 6.1.

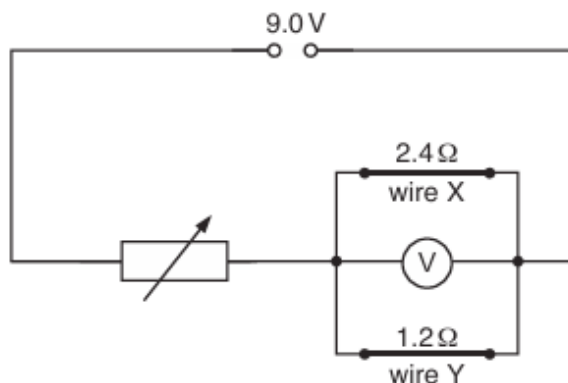


Fig. 6.1

Wire X has a resistance of $2.4\ \Omega$ and wire Y has a resistance of $1.2\ \Omega$. A voltmeter is connected in parallel with the wires. A variable resistor is used to adjust the power dissipated in wires X and Y.

The variable resistor is adjusted so that the voltmeter reads 6.0 V.

(i) Calculate the resistance of the variable resistor.

resistance = Ω [3]

(ii) Calculate the power dissipated in wire X.

power = W [2]

- (iii) The cross-sectional area of wire X is three times the cross-sectional area of wire Y. Assume that the resistivity and the number density of free electrons for the metal of both wires are the same.

Determine the ratio

1. $\frac{\text{length of wire X}}{\text{length of wire Y}}$,

ratio = [2]

2. $\frac{\text{average drift velocity of free electrons in wire X}}{\text{average drift velocity of free electrons in wire Y}}$.

ratio = [2]

[Total: 11]

27. 9702/22/O/N/18/Q6

(a) Define the *volt*.

.....
[1]

(b) A battery of electromotive force (e.m.f.) 7.0 V and negligible internal resistance is connected in series with three components, as shown in Fig. 6.1.

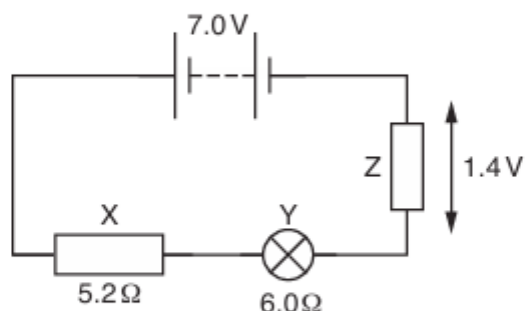


Fig. 6.1

Resistor X has a resistance of 5.2 Ω. The resistance of the filament wire of lamp Y is 6.0 Ω. The potential difference across resistor Z is 1.4 V.

(i) Calculate the current in the circuit.

current = A [2]

(ii) Determine the resistance of resistor Z.

resistance = Ω [1]

(iii) Calculate the percentage efficiency with which the battery supplies power to the lamp.

efficiency = % [3]

- (iv) The filament wire of the lamp is made of metal of resistivity $3.7 \times 10^{-7} \Omega \text{ m}$ at its operating temperature in the circuit.

Determine, for the filament wire, the value of α where

$$\alpha = \frac{\text{cross-sectional area}}{\text{length}} .$$

$$\alpha = \dots\dots\dots \text{ m [2]}$$

[Total: 9]

28. 9702/23/O/N/18/Q7

(a) State Kirchhoff's first law.

.....
[1]

(b) A potentiometer is connected to a battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance, as shown in Fig. 7.1.

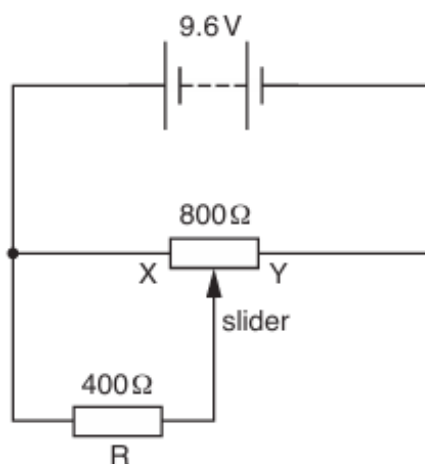


Fig. 7.1

The maximum resistance of the potentiometer is $800\ \Omega$. A resistor R of resistance $400\ \Omega$ is connected between the slider and end X of the potentiometer.

(i) State the potential difference across resistor R when the slider is positioned

1. at end X of the potentiometer,

potential difference = V

2. at end Y of the potentiometer.

potential difference = V
 [2]

- (ii) Calculate the potential difference across resistor R when the slider is positioned half-way between X and Y.

potential difference = V [3]

[Total: 6]

29. 9702/21/M/J/19/Q6

A battery of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor of resistance R , as shown in Fig. 6.1.

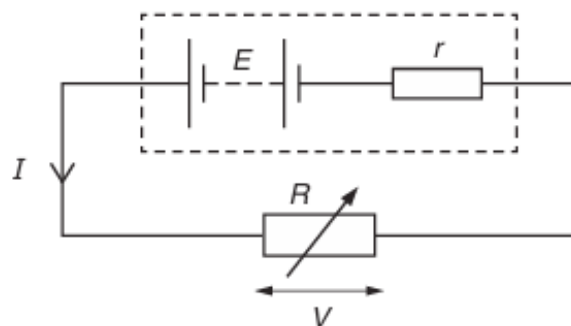


Fig. 6.1

The current in the circuit is I and the potential difference across the variable resistor is V .

- (a) Explain, in terms of energy, why V is less than E .

.....
[1]

- (b) State an equation relating E , I , r and V .

.....[1]

- (c) The resistance R of the variable resistor is varied. The variation with I of V is shown in Fig. 6.2.

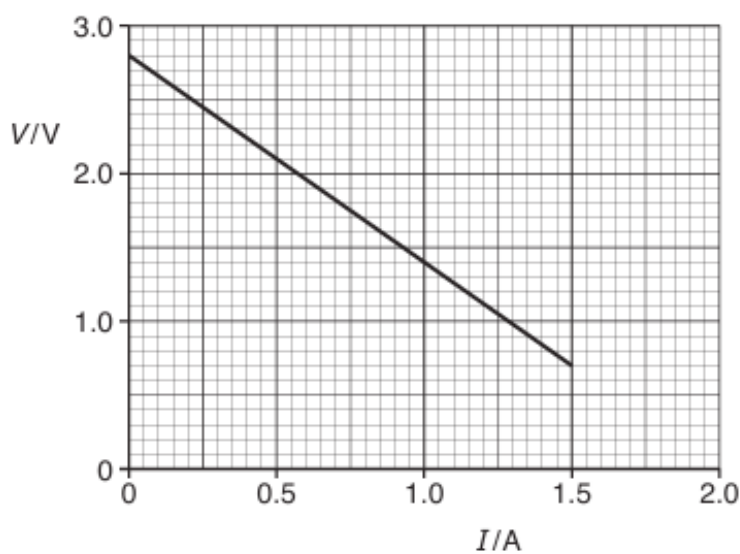


Fig. 6.2

Use Fig. 6.2 to:

- (i) explain how it may be deduced that the e.m.f. of the battery is 2.8 V

.....
.....[1]

- (ii) calculate the internal resistance r .

$r =$ Ω [2]

- (d) The battery stores 9.2 kJ of energy. The variable resistor is adjusted so that $V = 2.1$ V. Use Fig. 6.2 to:

- (i) calculate resistance R

$R =$ Ω [1]

- (ii) calculate the number of conduction electrons moving through the battery in a time of 1.0 s

number = [1]

- (iii) determine the time taken for the energy in the battery to become equal to 1.6 kJ.
(Assume that the e.m.f. of the battery and the current in the battery remain constant.)

time taken = s [3]

[Total: 10]

30. 9702/22/M/J/19/Q5

(a) State Kirchhoff's second law.

.....
[2]

(b) A battery of electromotive force (e.m.f.) 5.6 V and internal resistance r is connected to two external resistors, as shown in Fig. 5.1.

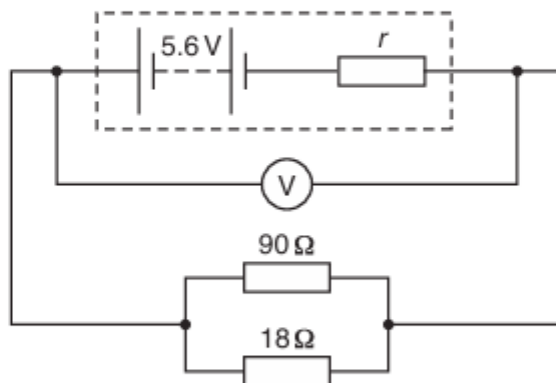


Fig. 5.1

The reading on the voltmeter is 4.8 V .

(i) Calculate:

1. the combined resistance of the two resistors connected in parallel

combined resistance = Ω [2]

2. the current in the battery.

current = A [2]

(ii) Show that the internal resistance r is $2.5\ \Omega$.

(iii) Determine the ratio

$$\frac{\text{power dissipated by internal resistance } r}{\text{total power produced by battery}}.$$

ratio = [3]

(c) The battery in (b) is now connected to a battery of e.m.f. 7.2V and internal resistance 3.5Ω. The new circuit is shown in Fig. 5.2.

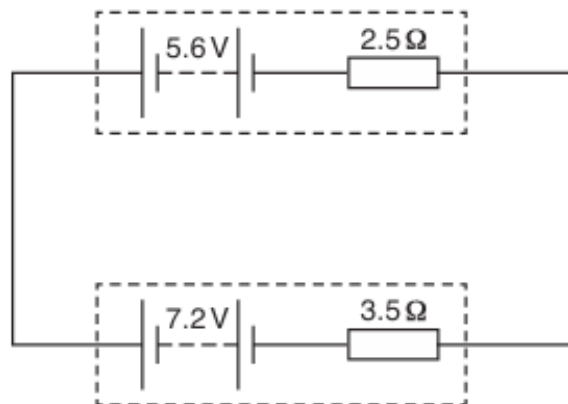


Fig. 5.2

Determine the current in the circuit.

current = A [2]

[Total: 13]

31. 9702/21/O/N/19/Q6

(a) Define *electric potential difference* (*p.d.*).

.....
..... [1]

(b) The variation with potential difference V of the current I in a semiconductor diode is shown in Fig. 6.1.

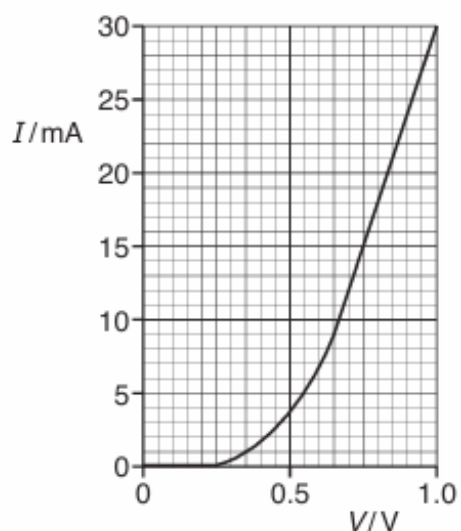


Fig. 6.1

Use Fig. 6.1 to describe qualitatively the variation of the resistance of the diode as V increases from 0 to 1.0 V.

.....
.....
.....
..... [2]

(c) The diode in (b) is part of the circuit shown in Fig. 6.2.

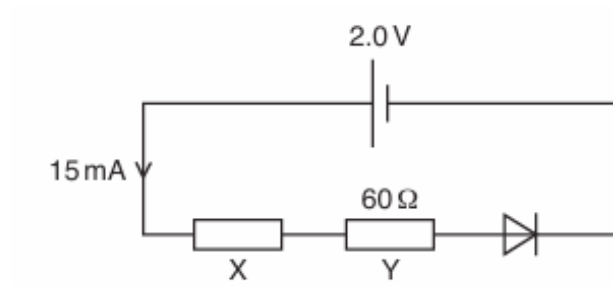


Fig. 6.2

The cell of electromotive force (e.m.f.) 2.0 V and negligible internal resistance is connected in series with the diode and resistors X and Y. The resistance of Y is $60\ \Omega$. The current in the cell is 15 mA.

(i) Use Fig. 6.1 to determine the resistance of the diode.

resistance = Ω [3]

(ii) Calculate:

1. the resistance of X

resistance = Ω [3]

2. the ratio

$$\frac{\text{power dissipated in resistor Y}}{\text{total power produced by the cell}}$$

ratio = [2]

32. 9702/22/O/N/19/Q6

(a) State Kirchhoff's first law.

.....
 [1]

(b) The variations with potential difference V of the current I for a resistor X and for a semiconductor diode are shown in Fig. 6.1.

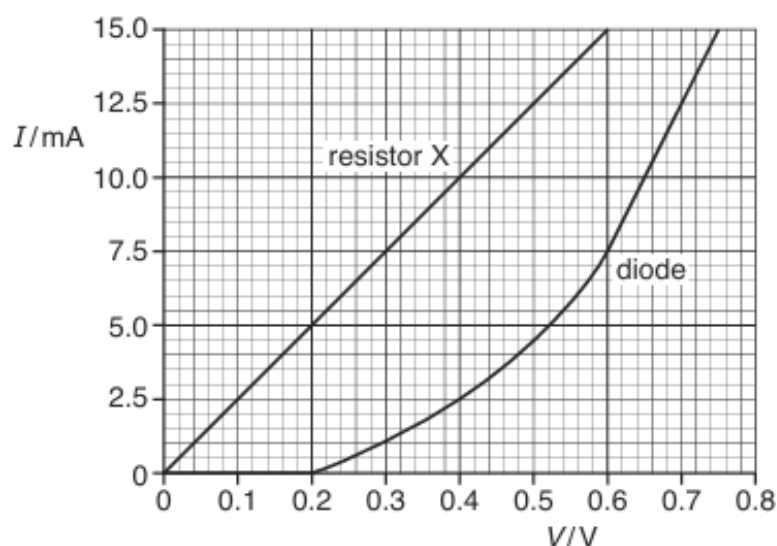


Fig. 6.1

(i) Determine the resistance of the diode for a potential difference V of 0.60 V.

resistance = Ω [3]

(ii) Describe, qualitatively, the variation of the resistance of the diode as V increases from 0.60 V to 0.75 V.

..... [1]

- (c) The diode and the resistor X in (b) are connected into the circuit shown in Fig. 6.2.

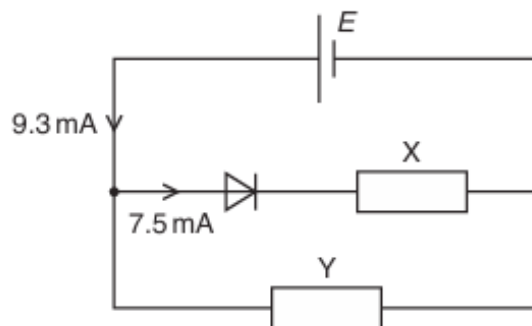


Fig. 6.2

The cell has electromotive force (e.m.f.) E and negligible internal resistance. Resistor Y is connected in parallel with resistor X and the diode. The current in the cell is 9.3 mA and the current in the diode is 7.5 mA .

- (i) Use Fig. 6.1 to determine E .

$$E = \dots\dots\dots \text{V} \quad [1]$$

- (ii) Determine the resistance of resistor Y .

$$\text{resistance} = \dots\dots\dots \Omega \quad [2]$$

- (iii) Calculate the power dissipated in the diode.

$$\text{power} = \dots\dots\dots \text{W} \quad [2]$$

- (iv) The cell is now replaced by a new cell of e.m.f. 0.50 V and negligible internal resistance. Use Fig. 6.1 to determine the new current in the diode.

$$\text{current} = \dots\dots\dots \text{mA} \quad [1]$$

33. 9702/23/O/N/19/Q6

A battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected to a network of two lamps and two resistors, as shown in Fig. 6.1.

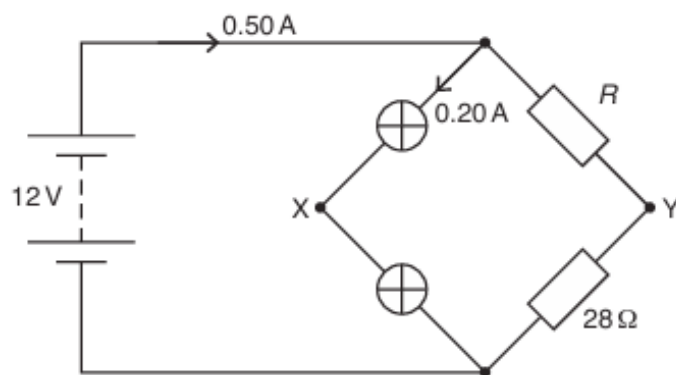


Fig. 6.1

The two lamps in the circuit have equal resistances. The two resistors have resistances R and $28\ \Omega$. The lamps are connected at junction X and the resistors are connected at junction Y. The current in the battery is 0.50 A and the current in the lamps is 0.20 A .

(a) Calculate:

(i) the resistance of each lamp

resistance = Ω [2]

(ii) resistance R .

$R =$ Ω [2]

(b) Determine the potential difference V_{XY} between points X and Y.

$V_{XY} =$ V [3]

(c) Calculate the ratio

$$\frac{\text{total power dissipated by the lamps}}{\text{total power produced by the battery}}.$$

ratio = [2]

(d) The resistor of resistance R is now replaced by another resistor of lower resistance.

State and explain the effect, if any, of this change on the ratio in (c).

.....
.....
.....
.....
..... [2]

[Total: 11]

34. 9702/21/M/J/20/Q5

(a) Metal wire is used to connect a power supply to a lamp. The wire has a total resistance of $3.4\ \Omega$ and the metal has a resistivity of $2.6 \times 10^{-8}\ \Omega\text{m}$. The total length of the wire is 59 m.

(i) Show that the wire has a cross-sectional area of $4.5 \times 10^{-7}\ \text{m}^2$.

[2]

(ii) The potential difference across the total length of wire is 1.8 V.

Calculate the current in the wire.

current = A [1]

(iii) The number density of the free electrons in the wire is $6.1 \times 10^{28}\ \text{m}^{-3}$.

Calculate the average drift speed of the free electrons in the wire.

average drift speed = ms^{-1} [2]

(b) A different wire carries a current. This wire has a part that is thinner than the rest of the wire, as shown in Fig. 5.1.

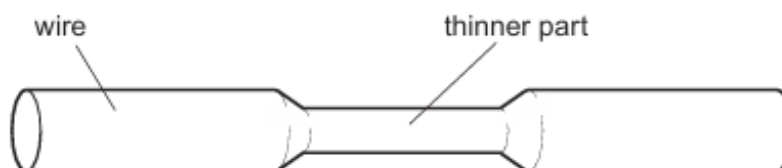


Fig. 5.1

- (i) State and explain qualitatively how the average drift speed of the free electrons in the thinner part compares with that in the rest of the wire.

.....

 [2]

- (ii) State and explain whether the power dissipated in the thinner part is the same, less or more than the power dissipated in an equal length of the rest of the wire.

.....

 [2]

- (c) Three resistors have resistances of $180\ \Omega$, $90\ \Omega$ and $30\ \Omega$.

- (i) Sketch a diagram showing how **two** of these three resistors may be connected together to give a combined resistance of $60\ \Omega$ between the terminals shown. Ensure you label the values of the resistances in your diagram.



[1]

- (ii) A potential divider circuit is produced by connecting the three resistors to a battery of electromotive force (e.m.f.) 12 V and negligible internal resistance. The potential divider circuit provides an output potential difference V_{OUT} of 8.0 V . Fig. 5.2 shows the circuit diagram.

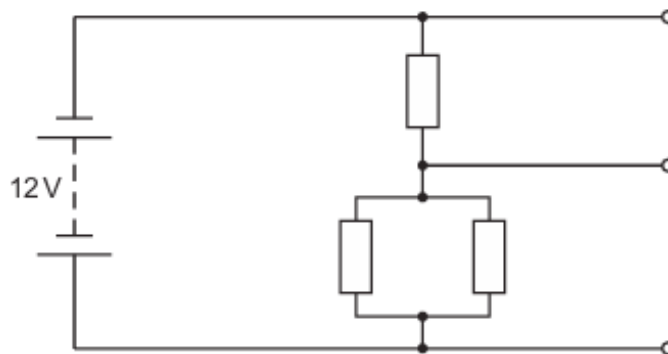


Fig. 5.2

On Fig. 5.2, label the resistances of all three resistors and the potential difference V_{OUT}

[2]

35. 9702/22/M/J/20/Q6

- (a) A battery of electromotive force (e.m.f.) 7.8 V and internal resistance r is connected to a filament lamp, as shown in Fig. 6.1.

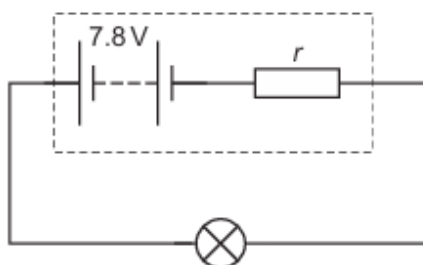


Fig. 6.1

A total charge of 750 C moves through the battery in a time interval of 1500 s . During this time the filament lamp dissipates 5.7 kJ of energy. The e.m.f. of the battery remains constant.

- (i) Explain, in terms of energy and without a calculation, why the potential difference across the lamp must be less than the e.m.f. of the battery.

.....
 [1]

- (ii) Calculate:

1. the current in the circuit

current = A [2]

2. the potential difference across the lamp

potential difference = V [2]

3. the internal resistance of the battery.

internal resistance = Ω [2]

(b) A student is provided with three resistors of resistances $90\ \Omega$, $45\ \Omega$ and $20\ \Omega$.

- (i) Sketch a circuit diagram showing how **two** of these three resistors may be connected together to give a combined resistance of $30\ \Omega$ between the terminals shown. Label the values of the resistances on your diagram.



[1]

- (ii) A potential divider circuit is produced by connecting the three resistors to a battery of e.m.f. 9.0V and negligible internal resistance. The potential divider circuit provides an output potential difference V_{OUT} of 3.6V . The circuit diagram is shown in Fig. 6.2.

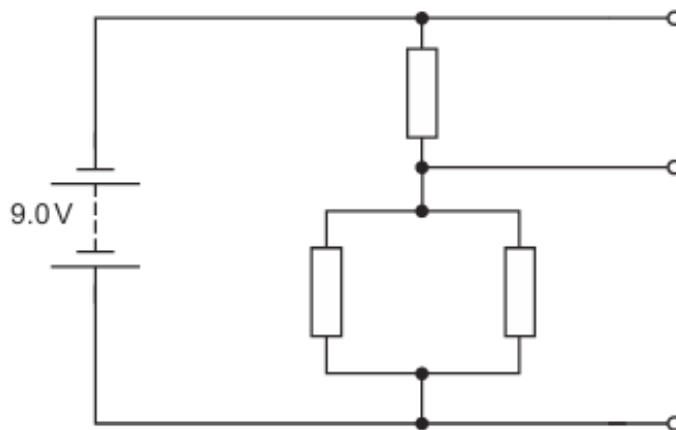


Fig. 6.2

On Fig. 6.2, label the resistances of all three resistors and the potential difference V_{OUT} . [2]

[Total: 10]

36. 9702/21/O/N/20/Q7

(a) Define the *ohm*.

.....
 [1]

(b) A uniform wire has resistance 3.2Ω . The wire has length 2.5m and is made from metal of resistivity $460\text{n}\Omega\text{m}$.

Calculate the cross-sectional area of the wire.

cross-sectional area = m^2 [3]

(c) A cell of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor of resistance R , as shown in Fig. 7.1.

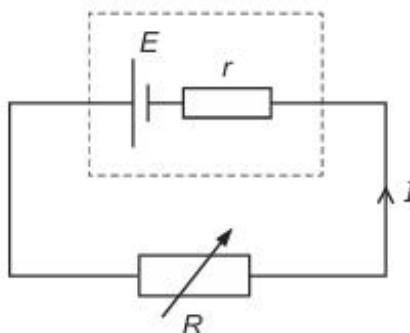


Fig. 7.1

The current in the circuit is I .

(i) State, in terms of energy, why the potential difference across the variable resistor is less than the e.m.f. of the cell.

.....
 [1]

- (ii) State an expression for E in terms of I , R and r .

$$E = \dots\dots\dots [1]$$

- (iii) The resistance R of the variable resistor is changed so that it is equal to r .

Determine an expression, in terms of only E and r , for the power P dissipated in the variable resistor.

$$P = \dots\dots\dots [2]$$

[Total: 8]

37. 9702/23/O/N/20/Q6

- (a) Define *electric potential difference* (p.d.).

.....
 [1]

- (b) A wire of cross-sectional area A is made from metal of resistivity ρ . The wire is extended. Assume that the volume V of the wire remains constant as it extends.

Show that the resistance R of the extending wire is inversely proportional to A^2 .

[2]

- (c) A battery of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor of resistance R , as shown in Fig. 6.1.

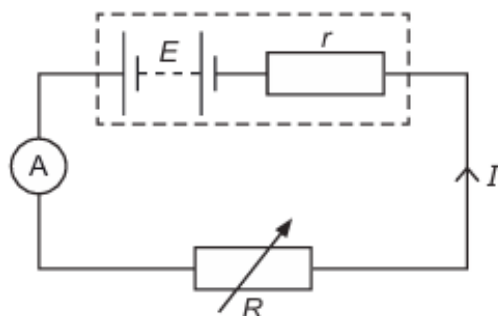


Fig. 6.1

The current in the circuit is I .

Use Kirchhoff's second law to show that

$$R = \left(\frac{E}{I} \right) - r.$$

[1]

- (d) An ammeter is used in the circuit in (c) to measure the current I as resistance R is varied. Fig. 6.2 is a graph of R against $\frac{1}{I}$.

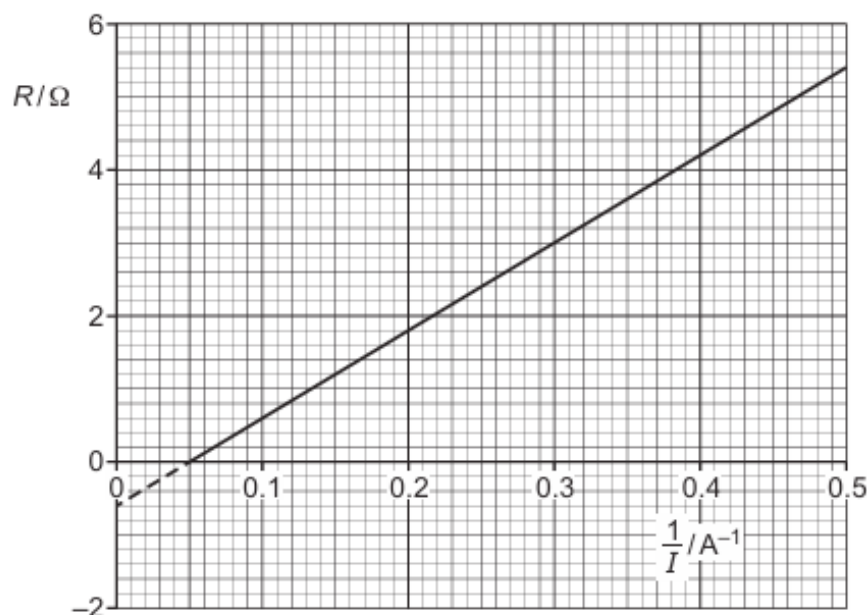


Fig. 6.2

- (i) Use Fig. 6.2 to determine the power dissipated in the variable resistor when there is a current of 2.0 A in the circuit.

power = W [3]

- (ii) Use Fig. 6.2 and the equation in (c) to:

1. state the internal resistance r of the battery

$r = \dots\dots\dots \Omega$

2. determine the e.m.f. E of the battery.

$E = \dots\dots\dots \text{ V}$
[3]

38. 9702/22/F/M/21/Q6

(a) State Kirchhoff's first law.

.....
 [1]

(b) A battery of electromotive force (e.m.f.) 12.0V and internal resistance r is connected to a filament lamp and a resistor, as shown in Fig. 6.1.

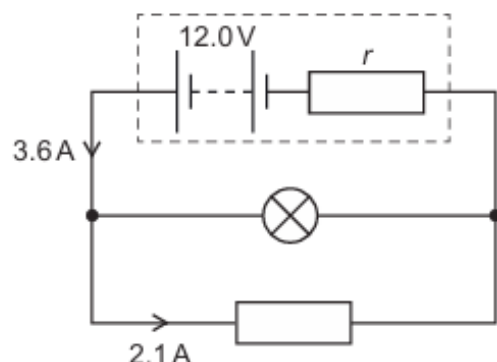


Fig. 6.1

The current in the battery is 3.6A and the current in the resistor is 2.1A. The I - V characteristic for the lamp is shown in Fig. 6.2.

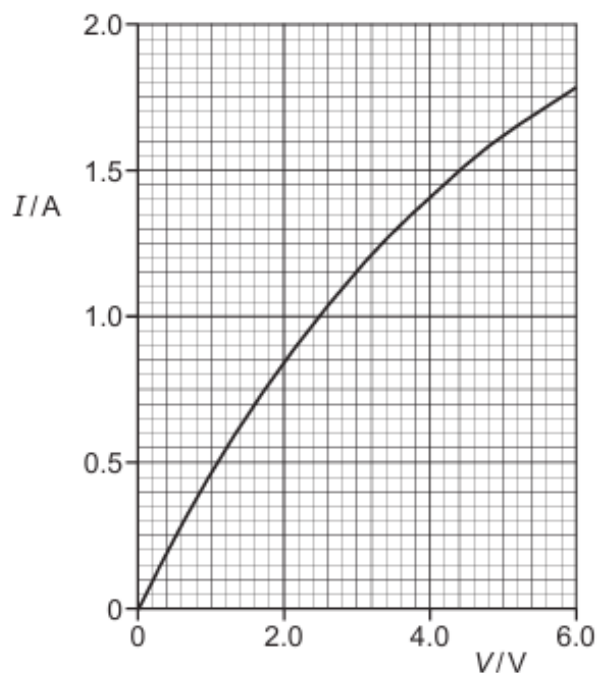


Fig. 6.2

- (i) Determine the resistance of the lamp in Fig. 6.1.

resistance = Ω [3]

- (ii) Determine the internal resistance r of the battery.

$r =$ Ω [2]

- (iii) The initial energy stored in the battery is 470 kJ. Assume that the e.m.f. and the current in the battery do not change with time.

Calculate the time taken for the energy stored in the battery to become 240 kJ.

time = s [2]

- (iv) The filament wire of the lamp is connected in series with the adjacent copper connecting wire of the circuit, as illustrated in Fig. 6.3.

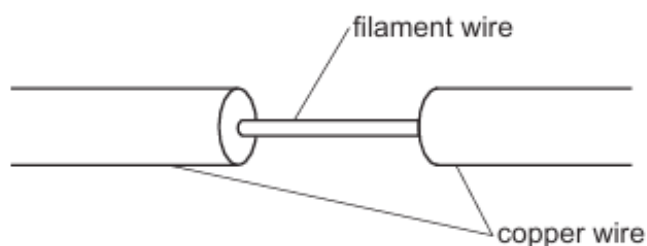


Fig. 6.3 (not to scale)

Some data for the filament wire and the adjacent copper connecting wire are given in Table 6.1.

Table 6.1

	filament wire	copper wire
cross-sectional area	A	$360A$
number density of free electrons	n	$2.5n$

Calculate the ratio

$$\frac{\text{average drift speed of free electrons in filament wire}}{\text{average drift speed of free electrons in copper wire}} .$$

ratio = [2]

[Total: 10]

39. 9702/21/M/J/21/Q5

(a) State Kirchhoff's second law.

.....

 [2]

(b) A battery has electromotive force (e.m.f.) 4.0 V and internal resistance $0.35\ \Omega$. The battery is connected to a uniform resistance wire XY and a fixed resistor of resistance R , as shown in Fig. 5.1.

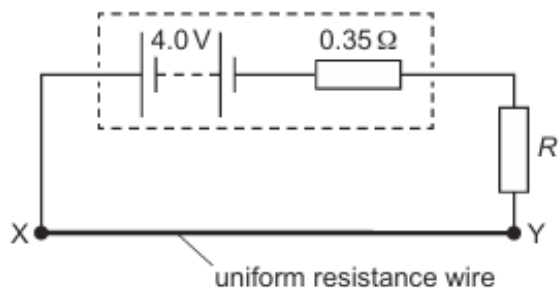


Fig. 5.1

Wire XY has resistance $0.90\ \Omega$. The potential difference across wire XY is 1.8 V .

Calculate:

(i) the current in wire XY

current = A [1]

(ii) the number of free electrons that pass a point in the battery in a time of 45 s

number = [2]

(iii) resistance R .

$R =$ Ω [2]

- (c) A cell of e.m.f. 1.2 V is connected to the circuit in (b), as shown in Fig. 5.2.

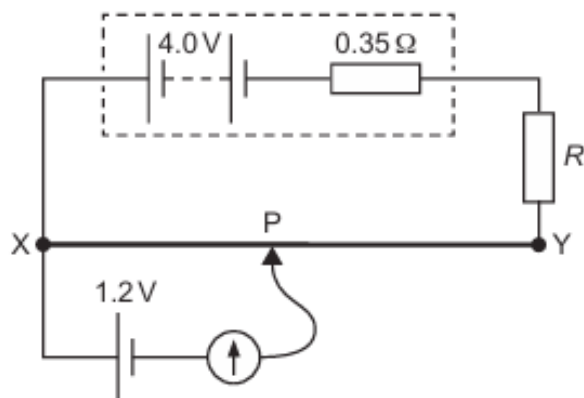


Fig. 5.2

The connection P is moved along the wire XY. The galvanometer reading is zero when distance XP is 0.30 m.

- (i) Calculate the total length L of wire XY.

$L = \dots\dots\dots$ m [2]

- (ii) The fixed resistor is replaced by a different fixed resistor of resistance greater than R .

State and explain the change, if any, that must be made to the position of P on wire XY so that the galvanometer reading is zero.

.....

 [2]

[Total: 11]

40. 9702/23/M/J/21/Q5

(a) Define the *electromotive force (e.m.f.)* of a source.

.....

.....

..... [2]

(b) The circuit shown in Fig. 5.1 contains a battery of e.m.f. E that has internal resistance r , a variable resistor, a voltmeter and an ammeter.

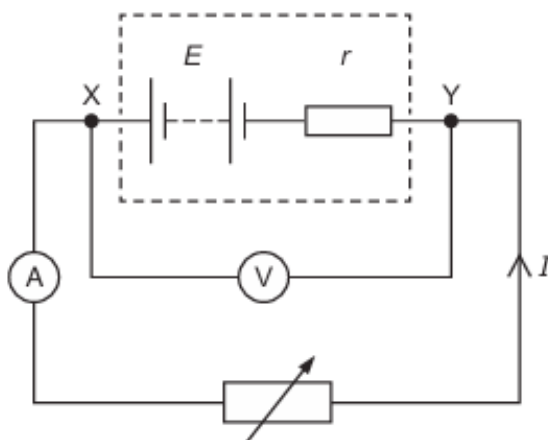


Fig. 5.1

Readings from the two meters are taken for different settings of the variable resistor. The variation with current I of the potential difference (p.d.) V across the terminals XY of the battery is shown in Fig. 5.2.

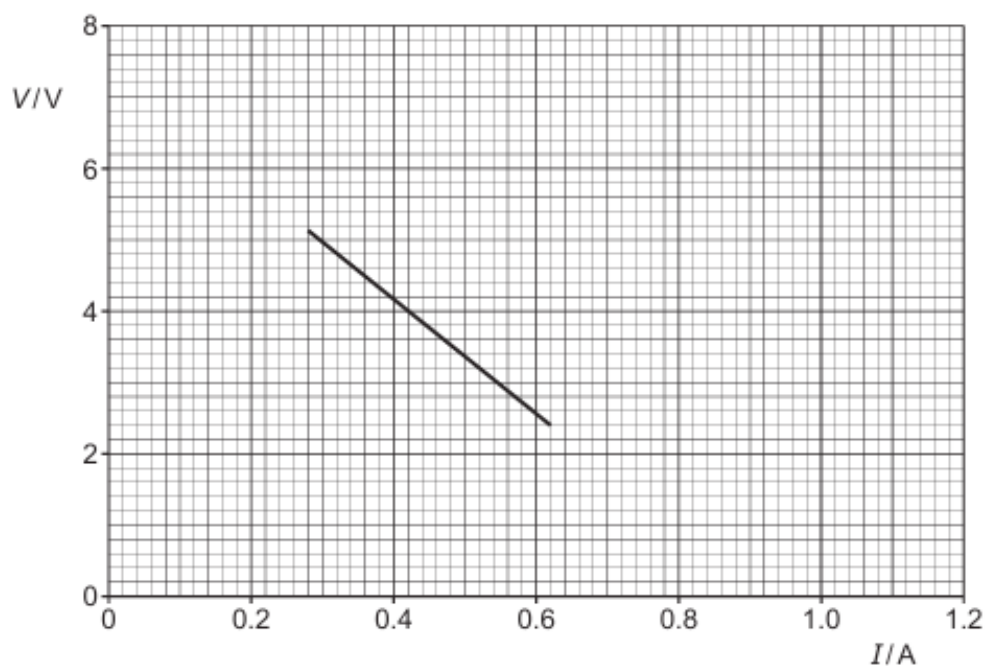


Fig. 5.2

Explain why V is not constant.

.....

.....

.....

..... [3]

(c) For the battery in (b), use Fig. 5.2 to determine:

(i) the e.m.f. E

$$E = \text{..... V [1]}$$

(ii) the maximum current that the battery can supply

$$\text{maximum current} = \text{..... A [1]}$$

(iii) the internal resistance r .

$$r = \text{..... } \Omega \text{ [2]}$$

(d) On Fig. 5.2, sketch a line to show a possible variation with I of V for a battery with a lower e.m.f. and a lower internal resistance than the battery in (b). Your line should extend over at least the same range of currents as the original line. [2]

[Total: 11]

41. 9702/21/O/N/21/Q5

(a) State Kirchhoff's first law.

.....

 [2]

(b) The circuit shown in Fig. 5.1 contains a battery of electromotive force (e.m.f.) E and negligible internal resistance connected to four resistors R_1 , R_2 , R_3 and R_4 , each of resistance R .

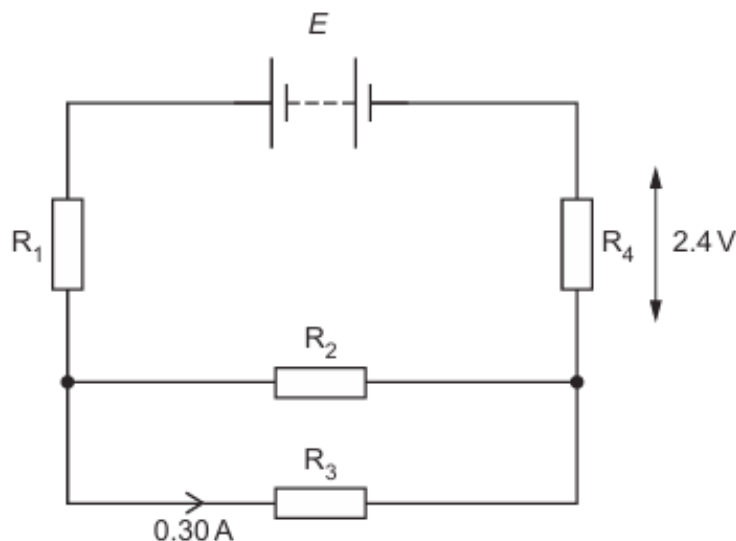


Fig. 5.1

The current in R_3 is 0.30 A and the potential difference (p.d.) across R_4 is 2.4 V .

(i) Show that R is equal to $4.0\ \Omega$.

[2]

(ii) Determine the e.m.f. E of the battery.

$E = \dots\dots\dots\text{ V}$ [2]

- (c) The battery in (b) is replaced with another battery of the same e.m.f. E but with an internal resistance that is not negligible.

State and explain the change, if any, in the total power produced by the battery.

.....
.....
..... [2]

- (d) The resistors in the circuit of Fig. 5.1 are made from nichrome wire of uniform radius $240\text{ }\mu\text{m}$. The length of this wire needed to make each resistor is 0.67 m .

Calculate the resistivity of nichrome.

resistivity = $\Omega\text{ m}$ [3]

[Total: 11]

42. 9702/23/O/N/21/Q6

- (a) A resistance wire of uniform cross-sectional area $3.3 \times 10^{-7} \text{ m}^2$ and length 2.0 m is made of metal of resistivity $5.0 \times 10^{-7} \Omega \text{ m}$.

Show that the resistance of the wire is 3.0Ω .

[2]

- (b) The ends of the resistance wire in (a) are connected to the terminals X and Y in the circuit shown in Fig. 6.1.

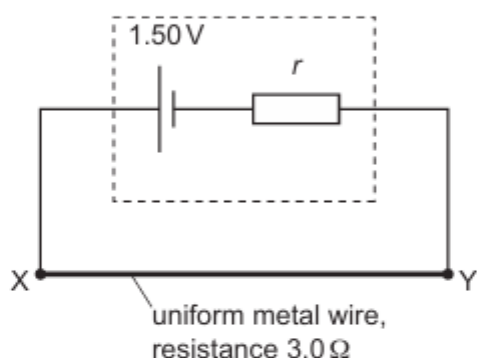


Fig. 6.1

The cell has an electromotive force (e.m.f.) of 1.50 V and internal resistance r . The potential difference between X and Y is 1.20 V .

Calculate:

- (i) the current in the circuit

current = A [1]

- (ii) the internal resistance r .

$r =$ Ω [2]

- (c) A galvanometer and a cell of e.m.f. E with negligible internal resistance are connected to the circuit in (b), as shown in Fig. 6.2.

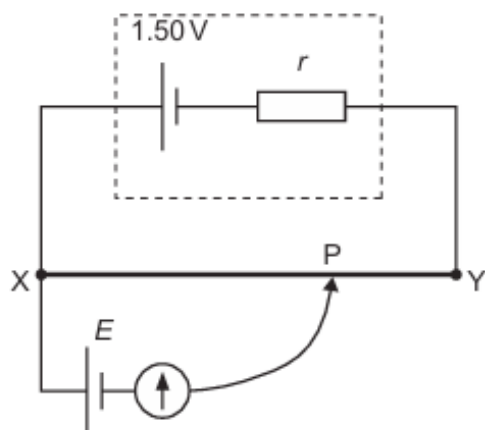


Fig. 6.2

The resistance wire between X and Y has a length of 2.0 m. The galvanometer has a reading of zero when the connection P is adjusted so that the length XP is 1.4 m.

Determine the e.m.f. E of the cell.

$$E = \dots\dots\dots \text{ V [2]}$$

- (d) The circuit in Fig. 6.2 is modified by replacing the original resistance wire with a second resistance wire. The second wire has the same length as the original wire and is made of the same metal.

The second wire has a smaller cross-sectional area than the original wire.

Connection P is adjusted on the second wire so that the galvanometer has a reading of zero.

State and explain whether length XP for the second wire is shorter than, longer than or the same as length XP for the original wire when the galvanometer reading is zero.

.....

.....

.....

.....

..... [3]

43. 9702/21/M/J/22/Q6

(a) State Kirchhoff's first law.

.....
..... [1]

(b) A battery is connected to two resistors X and Y, as shown in Fig. 6.1.

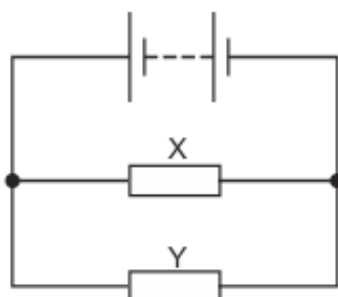


Fig. 6.1

The resistance of resistor X is greater than the resistance of resistor Y.

(i) State and explain which resistor dissipates more power.

.....
.....
.....
.....
..... [3]

(ii) The two resistors are made of wires that have the same length. Both wires are made from metal of the same resistivity.

State and explain which resistor is made of wire with the larger cross-sectional area.

.....
.....
..... [2]

- (c) A battery of electromotive force (e.m.f.) 9.0 V and negligible internal resistance is connected in series with a light-dependent resistor (LDR) and a fixed resistor of resistance $1800\ \Omega$, as shown in Fig. 6.2.

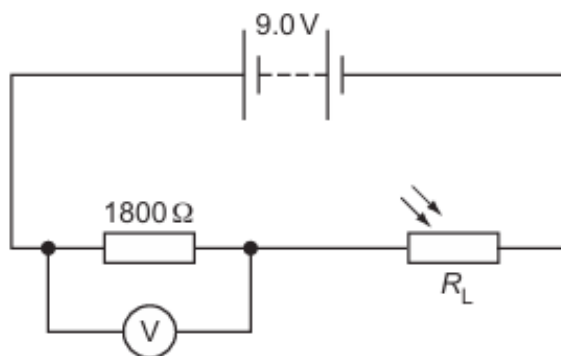


Fig. 6.2

A voltmeter is connected across the fixed resistor. The reading on the voltmeter is 5.4 V .

- (i) Calculate the current in the circuit.

current = A [1]

- (ii) Calculate the resistance R_L of the LDR.

R_L = Ω [2]

- (iii) The intensity of the light illuminating the LDR increases.

By reference to the current in the circuit, state and explain the change, if any, to the voltmeter reading.

.....

 [2]

[Total: 11]

44. 9702/21/O/N/22/Q5

(a) State Ohm's law.

.....

 [2]

(b) The variation of current I with potential difference V for a filament lamp is shown in Fig. 5.1.

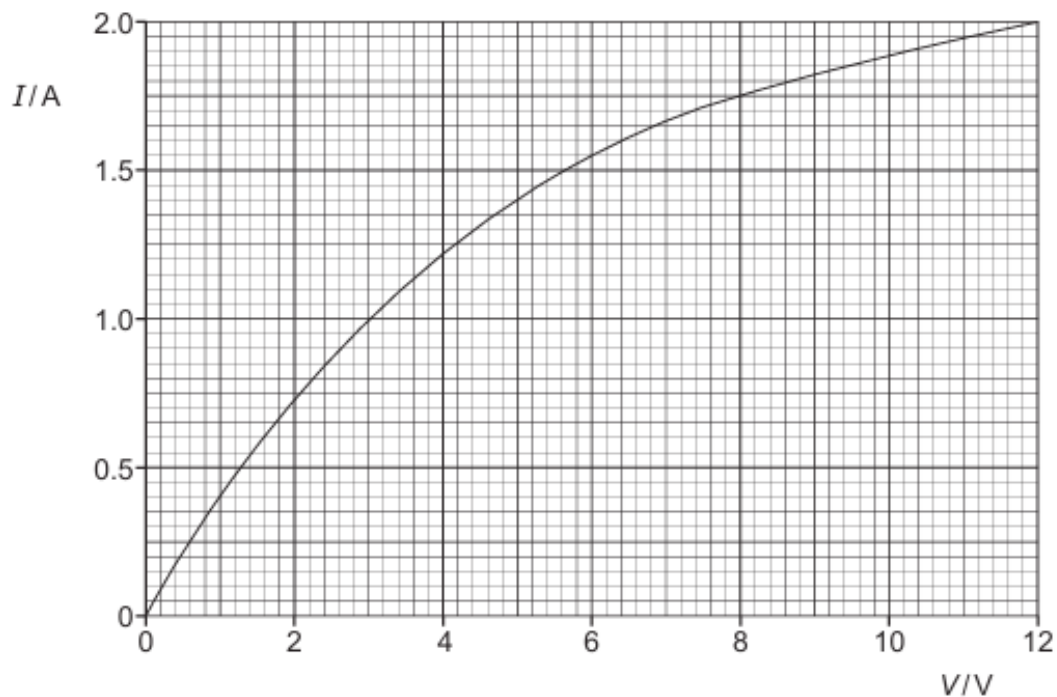


Fig. 5.1

The resistance of the filament lamp increases with potential difference.

(i) State how Fig. 5.1 shows this.

.....
 [1]

(ii) Explain why the resistance varies in this way.

.....
 [1]

- (c) Fig. 5.2 shows a circuit with a battery of electromotive force (e.m.f.) 12.0V connected to a linear potentiometer AB and two identical filament lamps P and Q.

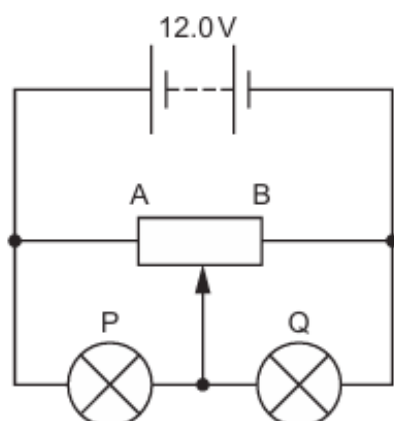


Fig. 5.2

The battery has negligible internal resistance and the lamps each have the same I – V characteristic shown in Fig. 5.1.

When the slider of the potentiometer is at its midpoint, as shown in Fig. 5.2, the current I in the battery is 1.78A.

Determine:

- (i) the current in lamp P

current = A [1]

- (ii) the total power dissipated in lamps P and Q

total power = W [2]

- (iii) the resistance of the potentiometer between its ends A and B.

resistance = Ω [2]

(d) The slider of the potentiometer in (c) is moved to end A.

State and explain the effect on the brightness of lamps P and Q.

lamp P:

.....

lamp Q:

.....

[2]

[Total: 11]

45. 9702/23/O/N/22/Q5

(a) State Kirchhoff's second law.

.....
.....
..... [2]

(b) Three identical cells, each of electromotive force (e.m.f.) 1.5 V and internal resistance $590\text{ m}\Omega$, are connected in parallel across a conductor, as shown in Fig. 5.1.

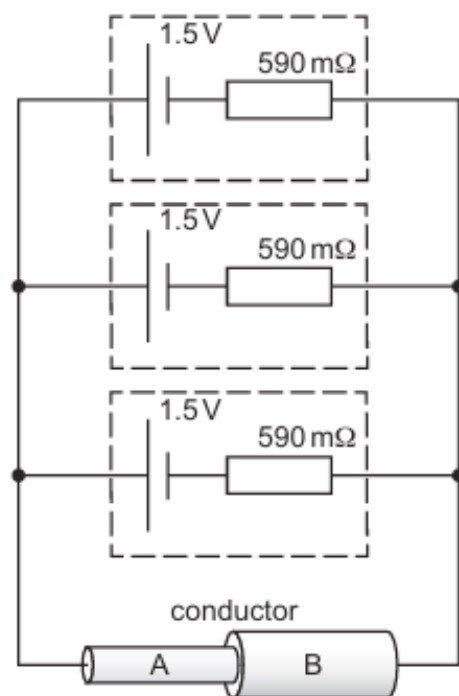


Fig. 5.1

The conductor is composed of two cylindrical sections A and B.
The total resistance of the circuit is $2.2\text{ }\Omega$.

(i) Show that the resistance of the conductor is $2.0\text{ }\Omega$.

- (ii) Calculate the current in the conductor.

current = A [2]

- (c) The two cylindrical sections A and B of the conductor in Fig. 5.1 are made from the same material and have the same length.

The diameter of section A is 4.3 mm and the diameter of section B is 7.6 mm.

The resistance of section A is R_A and the resistance of section B is R_B .

- (i) Calculate the ratio $\frac{R_A}{R_B}$.

$\frac{R_A}{R_B} = \dots\dots\dots$ [3]

- (ii) Calculate the ratio

$$\frac{\text{average drift speed of free electrons in section A}}{\text{average drift speed of free electrons in section B}}.$$

Explain your reasoning.

ratio = [2]

(d) The circuit of Fig. 5.1 is altered by removing one of the cells.

State and explain the effect, if any, of this change on the potential difference across the conductor.

.....

.....

.....

.....

..... [3]

[Total: 14]

46. 9702/21/M/J/23/Q7

A battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance is connected in series with two fixed resistors and a thermistor, as shown in Fig. 7.1.

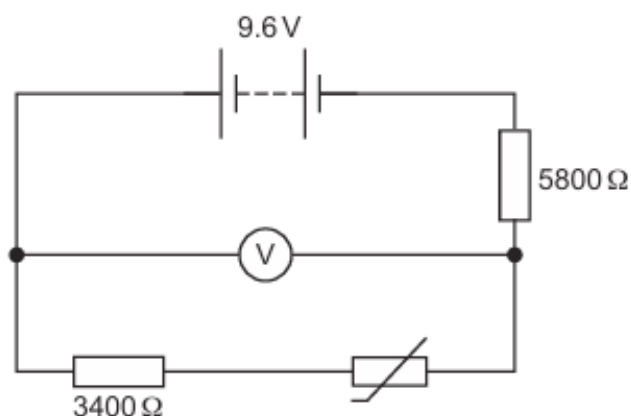


Fig. 7.1

The fixed resistors have resistances of 3400Ω and 5800Ω . The reading on the voltmeter in the circuit is 6.0 V .

- (a)** Calculate the current in the resistor of resistance 5800Ω .

current =A [2]

- (b)** Calculate the resistance of the thermistor.

resistance = Ω [2]

- (c) The initial energy stored in the battery is $2.6 \times 10^4 \text{ J}$.

Assume that the e.m.f. of the battery is constant.

Determine the final energy stored in the battery after a charge of 330 C has moved through it.

final stored energy = J [2]

- (d) The environmental conditions change causing an increase in the resistance of the thermistor.

State whether there is a decrease, increase or no change to:

- (i) the temperature of the thermistor

..... [1]

- (ii) the current in the thermistor

..... [1]

- (iii) the potential difference across the thermistor.

..... [1]

[Total: 9]

47. 9702/22/M/J/23/Q7

- (a) A battery of electromotive force (e.m.f.) 9.0V and negligible internal resistance is connected to a light-dependent resistor (LDR) and a fixed resistor, as shown in Fig. 7.1.

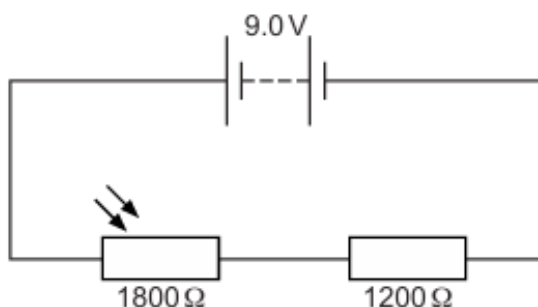


Fig. 7.1

The LDR and fixed resistor have resistances of $1800\ \Omega$ and $1200\ \Omega$ respectively.

Calculate the potential difference across the LDR.

potential difference = V [2]

- (b) The circuit in (a) is now modified by adding a uniform resistance wire XY and a galvanometer, as shown in Fig. 7.2.

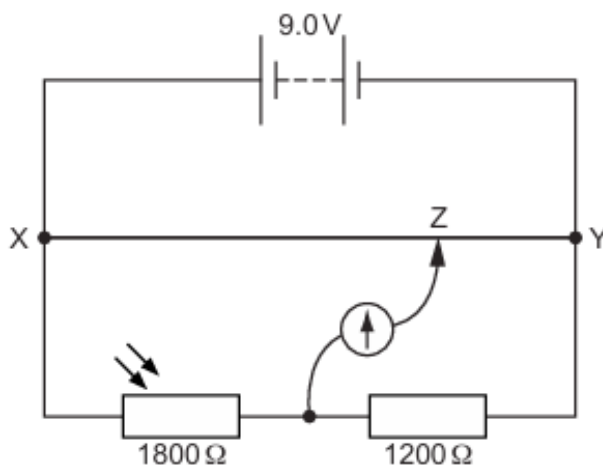


Fig. 7.2 (not to scale)

The length of the wire XY is 1.2m . The movable connection Z is positioned on the wire XY so that the galvanometer reading is zero.

- (i) Calculate the length XZ along the resistance wire.

length XZ = m [2]

- (ii) The environmental conditions change causing a decrease in the resistance of the LDR. The temperature of the LDR remains constant.

State whether there is a decrease, increase or no change to:

- the intensity of the light illuminating the LDR

.....

- the total power produced by the battery

.....

- the length XZ so that the galvanometer reads zero.

.....

[3]

[Total: 7]